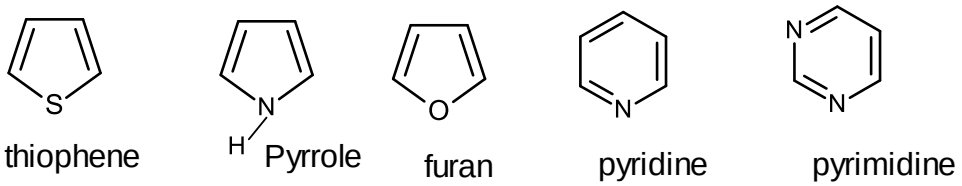
HETEROCYCLIC COMPOUNDS

Heterocyclic compounds are those cyclic compounds in which one or more ring carbon atoms are replaced by atoms of other elements [hetero atoms].

Common hetero atoms are N, O & S.

Examples:



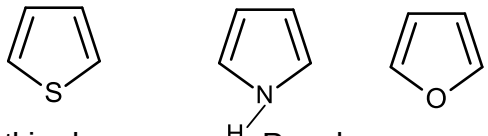
thiophene Pyrrole furan pyridine pyrimidine

Classification: Heterocyclic compounds are classified as follows:

1**. On the basis of size of the ring**: on the basis of size of the heterocyclic ring they are classified into two types

i) Five membered heterocycles: these contain 5 atoms in the ring (including hetero atom).

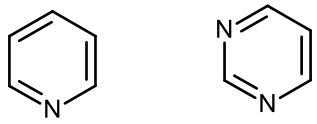
Examples:



thiophene Pyrrole furan

ii) Six membered heterocycles: These contain six atoms in the ring.

Examples:

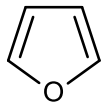


pyridine pyrimidine

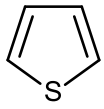
2. **Classification on the basis of the type of hetero atom**:

Heterocycles may be divided on the basis whether they contain O, N & S in the ring. This is regardless of the size of the ring.

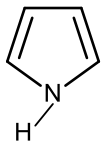
i) O- containing heterocycles: examples:

Furan 

ii) S-containing heterocycles: examples:

thiophene 

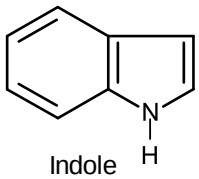
iii) N-containing heterocycles: examples:

Pyrrole 

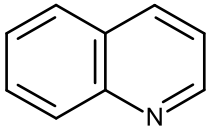
3. **CONDENSED RING HETEROCYCLES**:

These heterocycles are obtained by condensing heterocyclic ring with an aromatic ring. In these heterocycles the heterocyclic ring is fused with an aromatic ring.

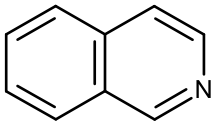
Examples:



Indole



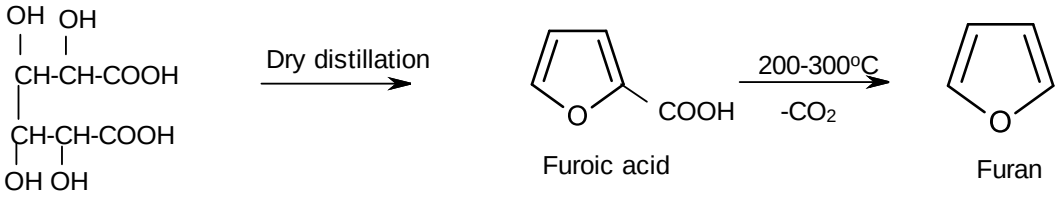
Quinoline



Isoquinoline

SYNTHESIS OF HETEROCYCLIC COMPOUNDS:

I**) SYNTHESIS OF FURAN**: Mucic acid on dry distillation yields furoic acid which undergoes decarboxylation to give furan when it is heated to 200-300°C

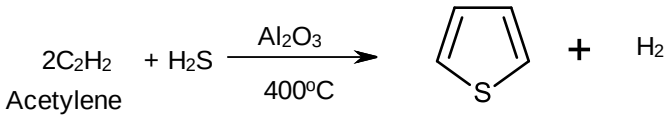


Mucic acid

Furoic acid

Furan

2**. SYNTHESIS OF THIOPHENE**: Thiophene is manufactured by passing a mixture of acetylene and H_2S through a tube containing Alumina at 400°C.

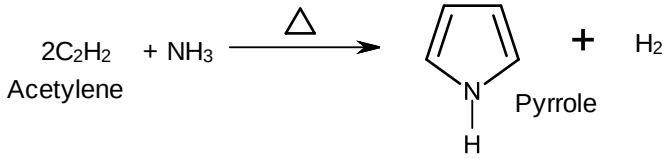


Acetylene

Thiophene

3. SYNTHESIS OF PYRROLE:

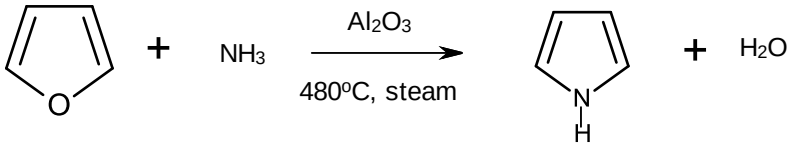
I. By passing a mixture of acetylene & NH_3 through a red-hot tube.



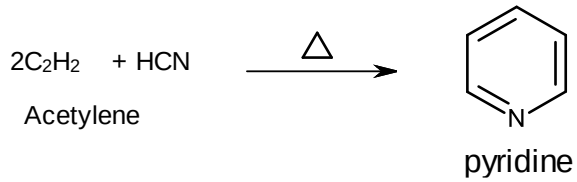
Acetylene

Pyrrole

II. By passing a mixture of furan, NH_3 and steam over alumina catalyst at a temperature of 480-490° C.

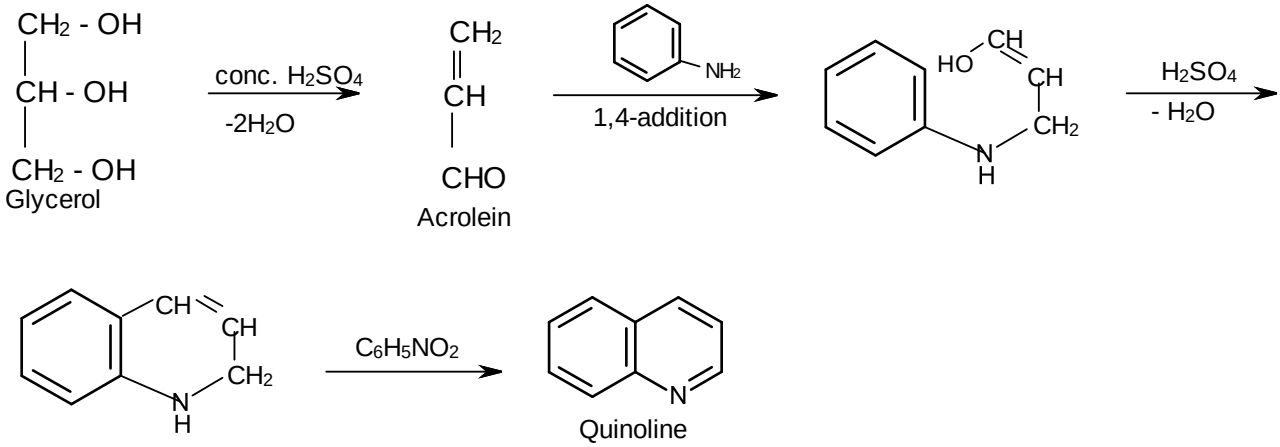


4. SYNTHESIS OF PYRIDINE: Pyridine is obtained by passing a mixture of acetylene & HCN through a red-hot tube

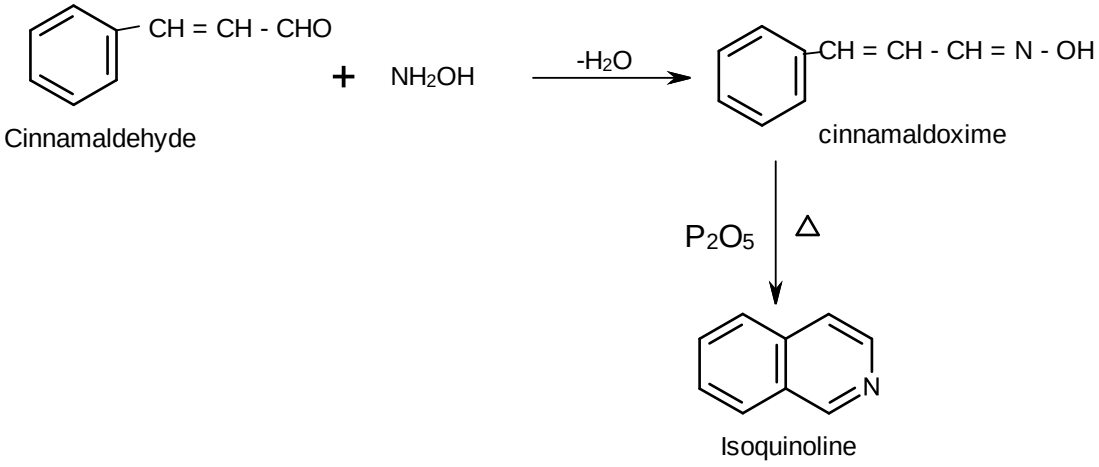


5**. SYNTHESIS OF QUINOLINE**: (**Skraup synthesis**)

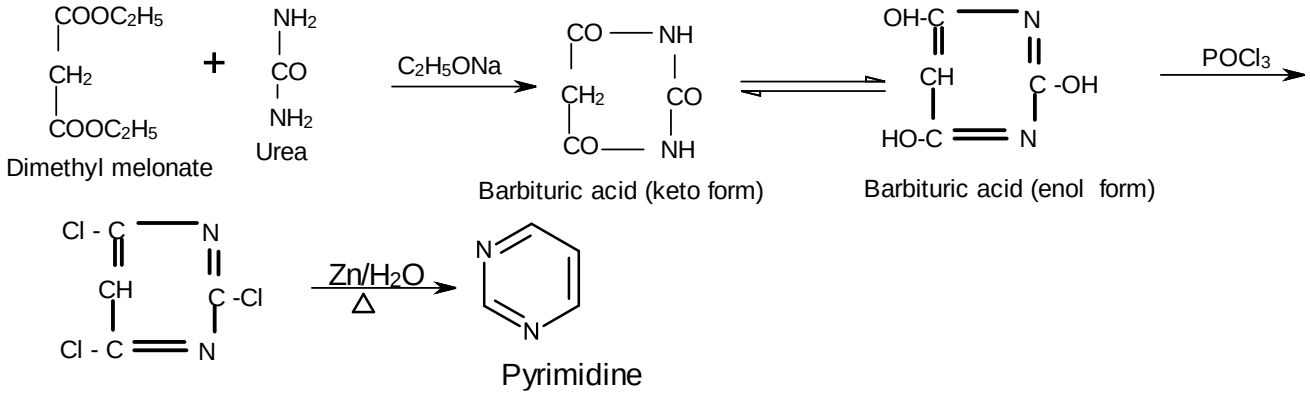
Quinoline is prepared by heating a mixture of aniline, glycerol and conc. H_2SO_4 in presence of a mild oxidizing agent such as nitro benzene or arsenic pentoxide.



6**. SYNTHESIS OF ISO QUINOLINE**: This involves condensation of cinnamaldehyde with hydroxylamine to form the corresponding oxime. The oxime is then heated with P_2O_5 to yield isoquinoline.



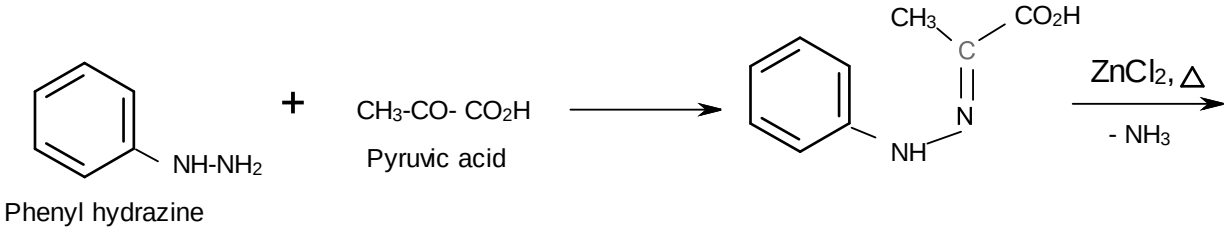
7. **SYNTHESIS OF PYRIMIDINE**: Urea is condensed with malonic ester by refluxing in alcoholic solution in presence of sodium ethoxide to form barbituric acid. Barbituric acid is then treated with $POCl_3$ (phosphorous oxy chloride) to form the trichloroderivative of barbituric acid which is then reduced by heating with zinc dust & water to give pyrimidine.

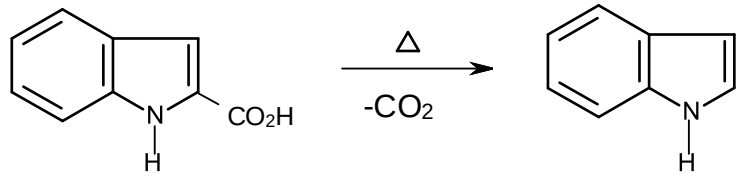


8. **SYNTHESIS OF INDOLE: By the Fischer indole synthesis**:

In this method pyruvic acid is treated with phenyl hydrazine to form the corresponding phenyl hydrazone.

The hydrazone is heated with anhydrous zinc chloride to give indole – 2- carboxylic acid which on decarboxylation yields indole.





Indole -2-carboxylic acid

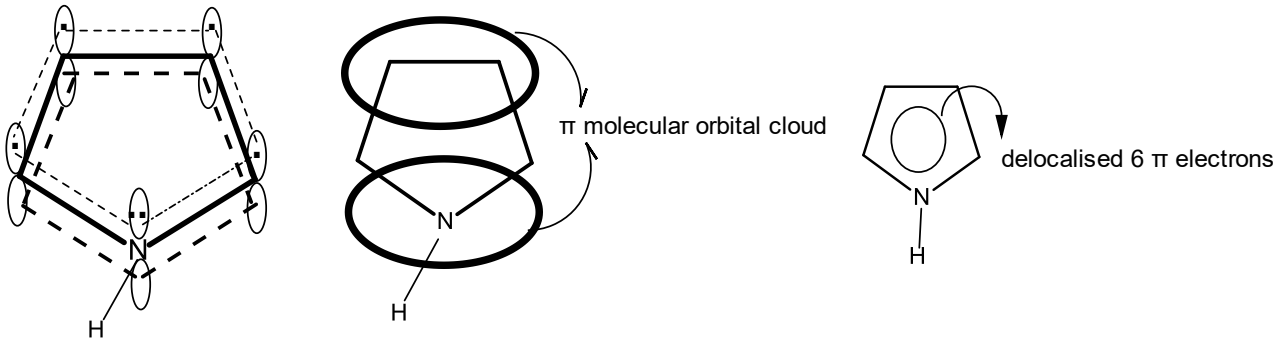
Indole

AROMATICITY OF PYRROLE AND PYRIDINE

i) Structure of pyrrole:

It can be explained by considering orbital structure. In pyrole all the 4 carbon atoms and one nitrogen atom are in sp^2 hybridised state. The sp^2 hybrid orbitals of these atoms overlap with each other and with 5 half filled 1s orbitals of 5 hydrogen atoms. Now each of the 4 carbon atoms has an unhybridised half filled p-orbital which lies at right angles to the plane of the pentagonal ring system. The p-orbital of nitrogen containing a lone pair of electrons also stands at right angles to the plane of the ring. These orbitals overlap sideways to form π molecular orbital clouds (one below & one above the plane of the ring) which contains total of six electrons (aromatic sextet) as in the case of benzene. The stabilization of the ring system due to

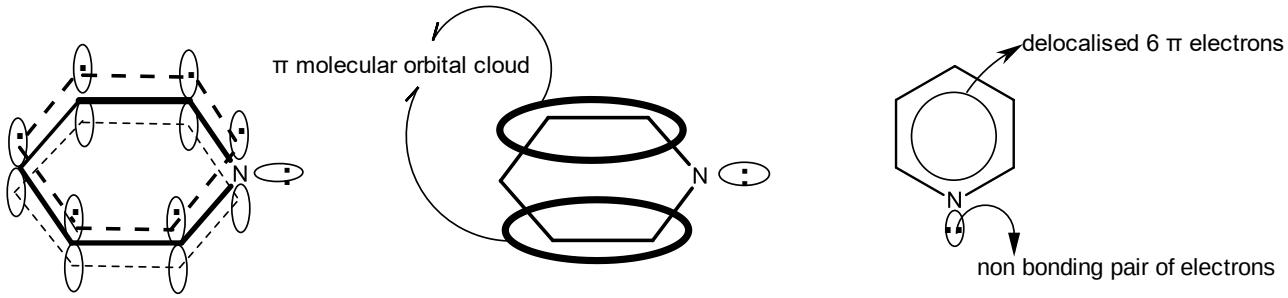
delocalization of 6 π electrons causes it to be aromatic. The resulting π molecular orbital satisfies the Huckel's rule (n= 1 in 4n+2) which is characteristic of aromatic compounds.



STRUCTURE OF PYRIDINE

5 carbon atoms and nitrogen atom of pyridine ring are sp^2 hybridised. Two of the sp^2 hybridised orbitals on each atom overlap with each other to form the C-C and C-N bonds. The third sp^2 orbital on each carbon atom overlaps with s-orbitals from hydrogen atoms to form the C-H bonds. The third sp^2 orbital of the nitrogen atom containing a lone pair of electrons remains unused and lies in the plane of the ring.

Also each ring atom in pyridine possesses an unhybridised p-orbital (containing one electron each) and they are perpendicular to the plane of the ring. Lateral overlap of the p-orbitals produces the delocalized π - molecular orbital cloud (one below & one above the plane of the ring) containing 6 electrons (aromatic sextet). The stabilization energy due to delocalization of
π - electrons is almost the same as for the benzene. Thus, pyridine is as stable as benzene and exhibits aromatic character. The resulting π - molecular orbital satisfies the Huckel's rule. (n = 1 in 4n +2)



Why pyrole is weaker base than pyridine?

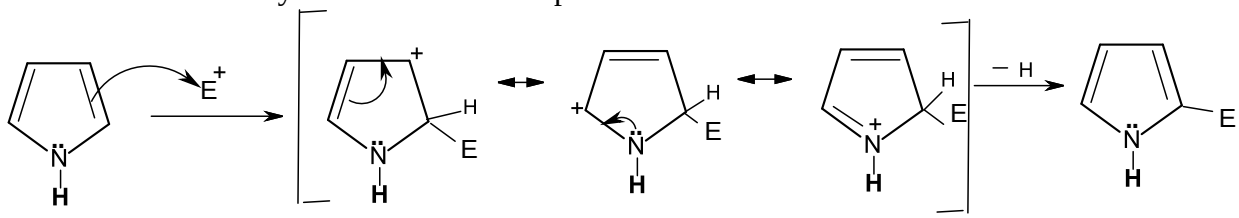
In pyrole, the lone pair of electrons in the N-atom is involved in the formation of aromatic sextet & hence less available for protonation whereas in the pyridine, the lone pair of electrons present in sp^2 hybrid orbitals of N atom is not involved in aromatic sextet and hence readily available for protonation. Therefore, pyrole is a weaker base whereas pyridine is a stronger base.

Electrophilic & nucleophilic substitution reactions of pyrrole & pyridine

Electrophilic substitution reactions of Pyrole:

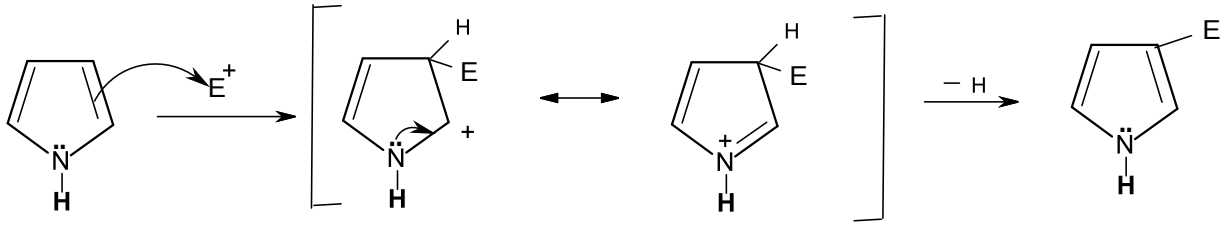
Pyrole undergoes electrophilic substitution reactions at C-2

This is because intermediate obtained by attack at C-2, is more stable with three resonance structures whereas only two such forms are possible for substitution at C-3.



(Three resonance structures - more stable)

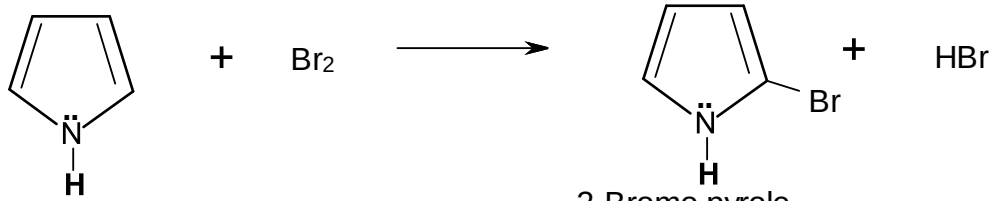
Attack at C – 3 position



(Two resonance structures - less stable)

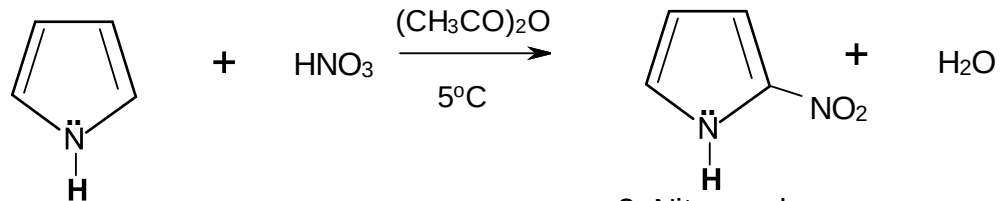
Examples:

a. Halogenation:



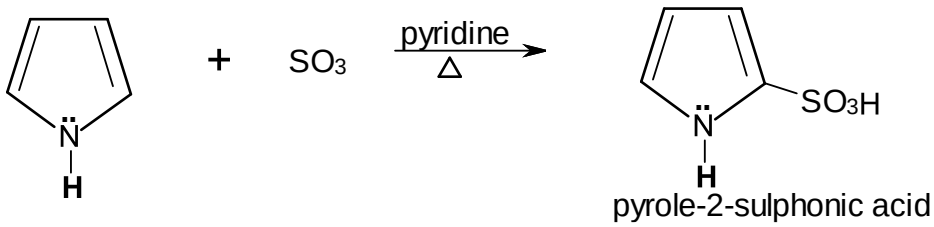
2-Bromo pyrole

b. Nitration:

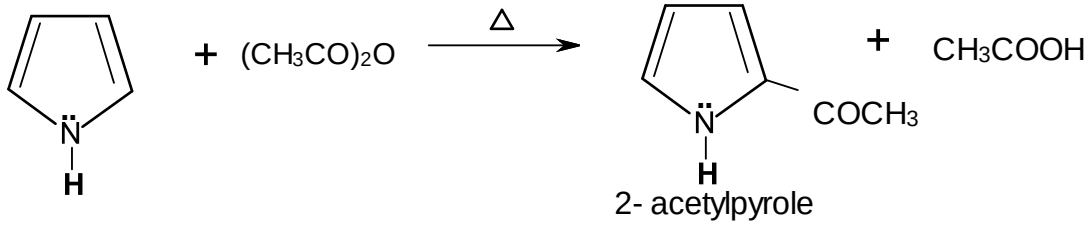


2- Nitropyrole

c. Sulphonation:



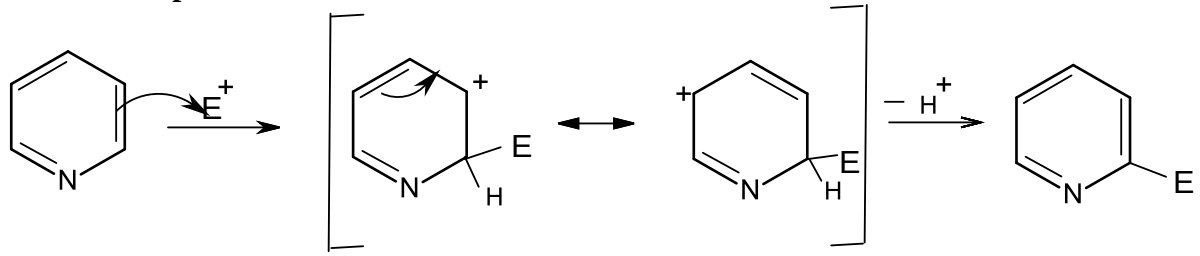
d. Friedel Craft's alkylation:



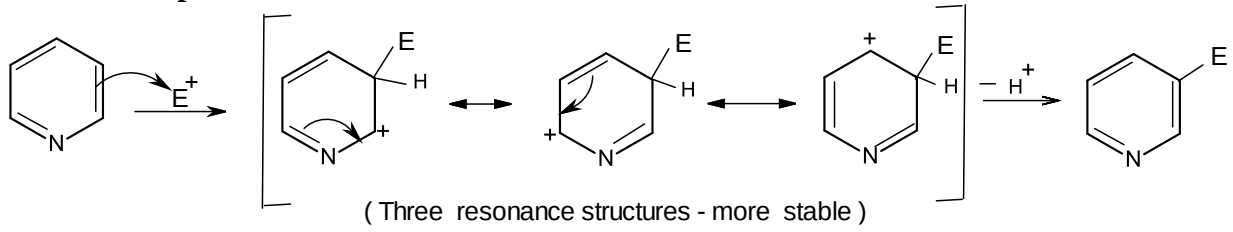
Electrophilic substitution reactions of pyridine:

Pyridine undergoes electrophilic substitution reactions at C – 3. this is because intermediate obtained by attack at C-3, is more stable with three resonance structures where as intermediate formed by attack at C- 2 or C-4 is less stable with two resonance structures.

Attack to C-2 postion:

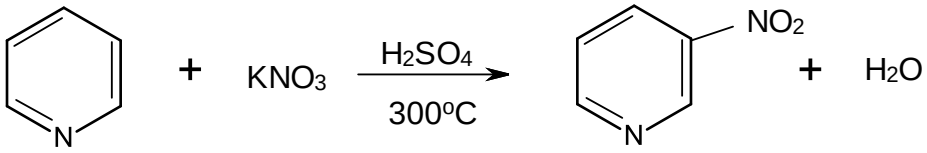


Attack at C-3 position:



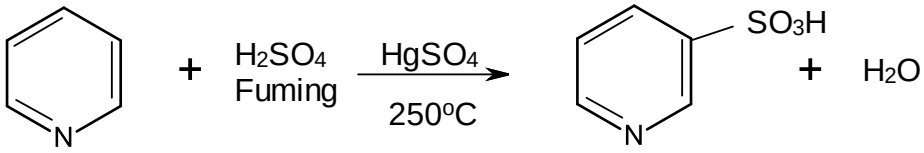
Examples for electrophilic substitution reactions:

a. Nitration:



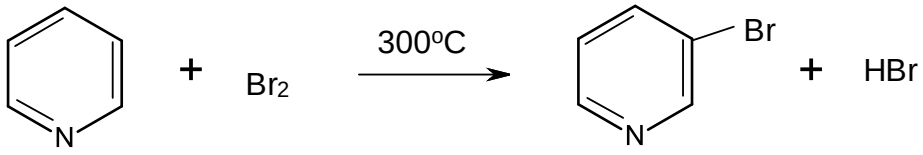
3-Nitropyridine

b. Sulphonation :



3-pyridinesulphonic acid

b. Bromination :



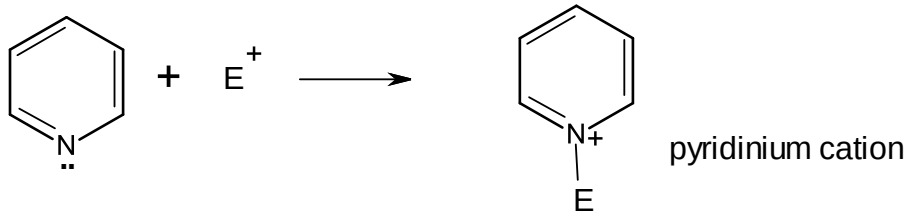
3-bromopyridine

Note:

Pyridine is less reactive than benzene towards electrophilic substitution reactions. This is because

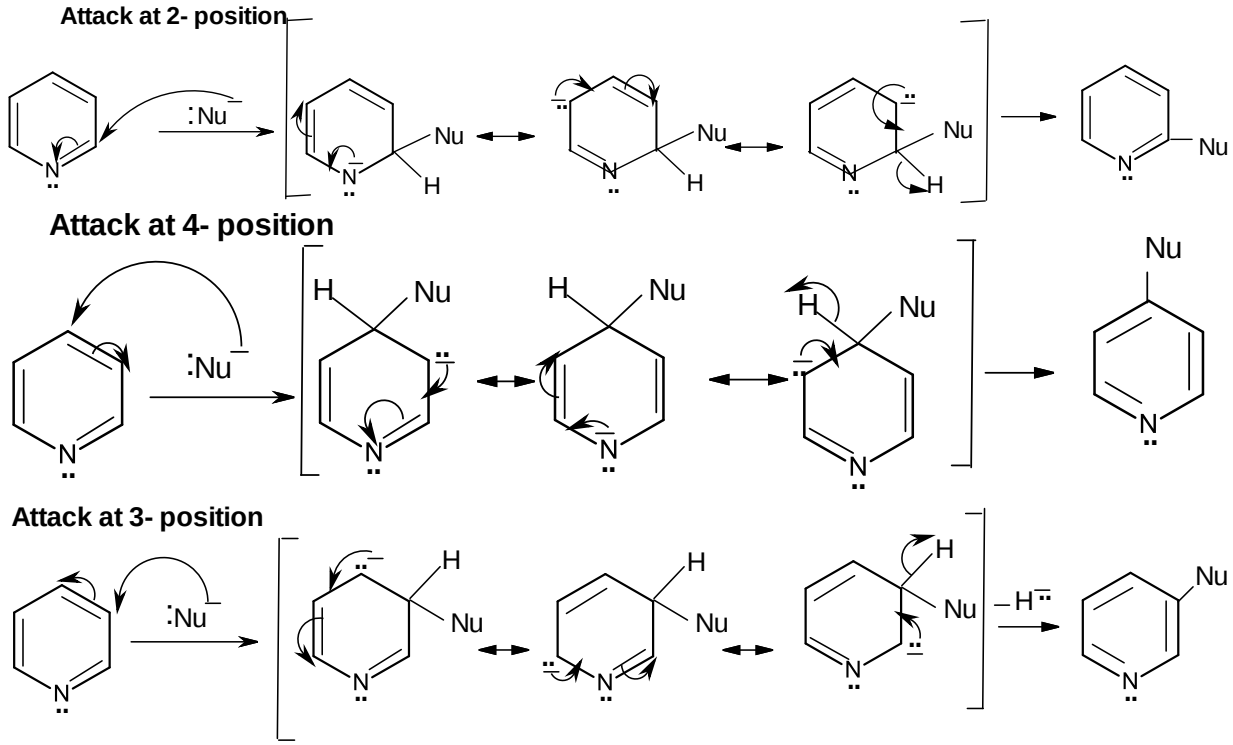
i) the nitrogen atom in pyridine being electronegative lowers the electron density around the ring carbons

ii) the electrophile coordinates with lone pair of electrons on nitrogen atom of pyridine to form resonance stabilized pyridinium salt



pyridinium cation

Nucleophilic substitution reactions: **Unlike benzene and pyrrole**, pyridine is highly reactive towards nucleophilic substitution reactions. Due to the high electronegativity of nitrogen atom of the pyridine ring which lowers the electron density around the pyridine ring carbons, facilitates the nucleophilic substitution reactions. Substitution takes place mainly at C-2 or C-4 than C-3 which can be understood by the stability of intermediate ion by writing resonance structures as given below.

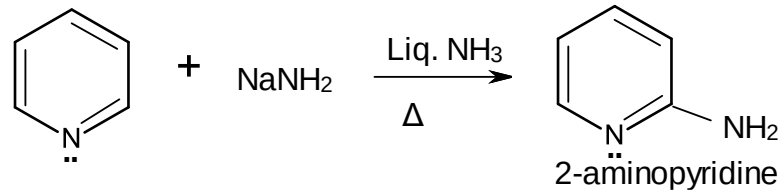


In the above intermediate forms attack at C-2 & C-4 involved intermediate in which negative charge is spread over nitrogen as well as carbons. This will give rise to enhanced stability to the intermediates. Consequently, the products with substituents at C-2 and C-4 predominates.

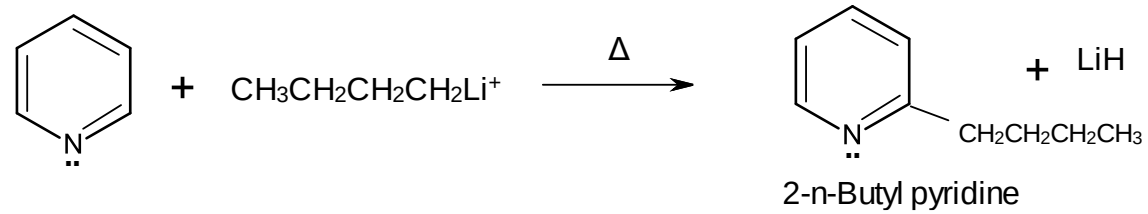
Examples:

1. Reaction with sodamide: (Chichibabin reaction)

Pyridine reacts with sodamide in liquid ammonia at about 100°C to form 2-amino pyridine.

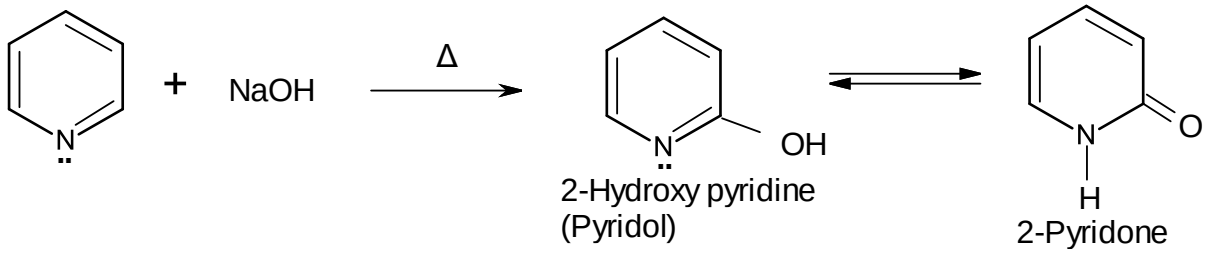


2. Reaction with n-butyl lithium: Pyridine reacts with n-butyllithium at 100°C to give 2-n-butyl lithium



3. Reaction with sodium hydroxide:

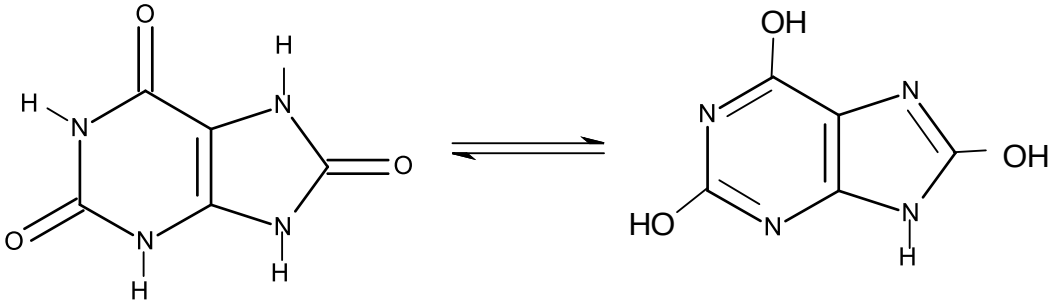
Pyridine reacts with sodium hydroxide at 300°C to yield an equilibrium mixture of 2-hydroxypyridine and 2-pyridone.



URIC ACID

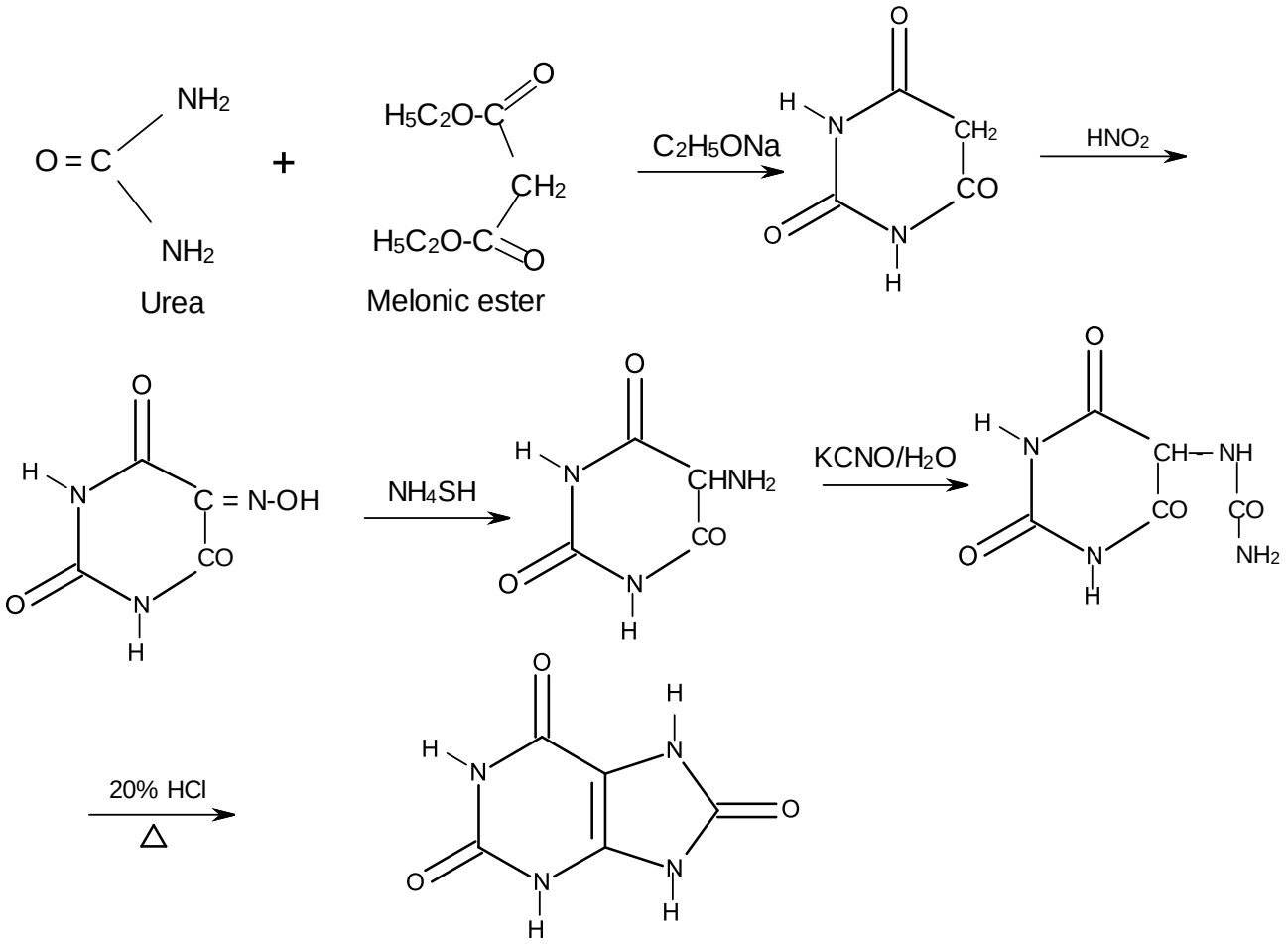
Uric acid is the chief constituent of the solid excrement of birds, reptiles and insects. It is so named as it was first isolated by Scheel and Bergmann from human urine.

Uric acid is a most important derivative of purine. It is a mixture of 2 tautomeric forms.



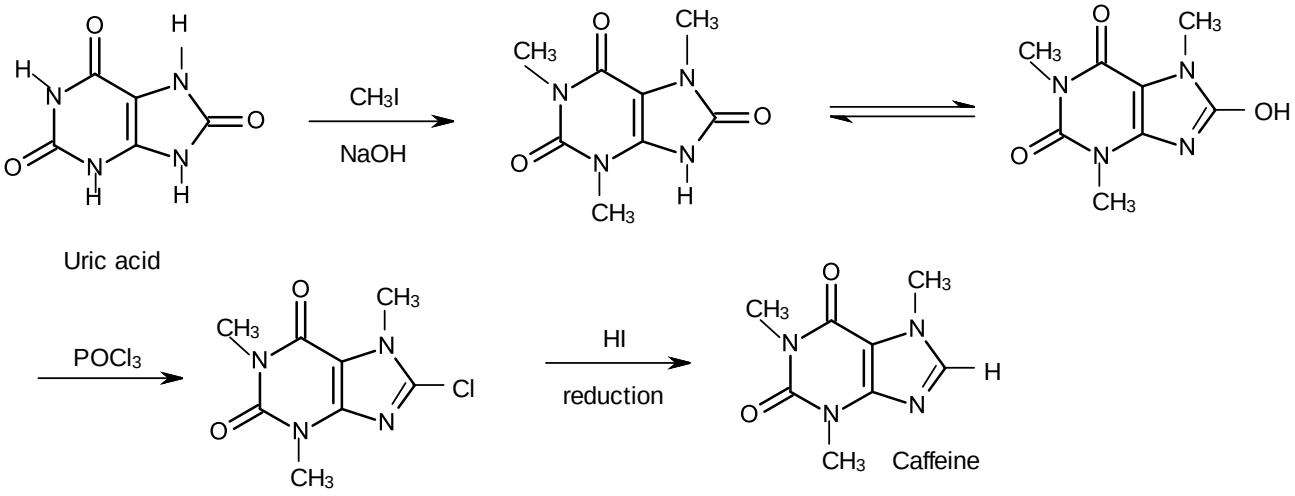
uric acid (keto form) uric acid (enol form)

Synthesis of uric acid (by Fischer's Synthesis):

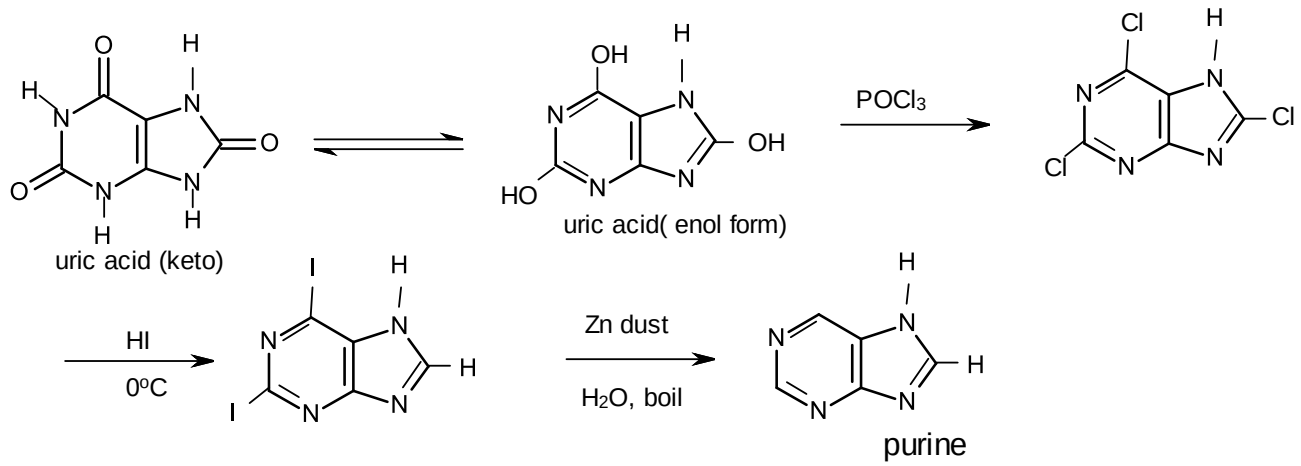


Conversion of uric acid to caffeine:

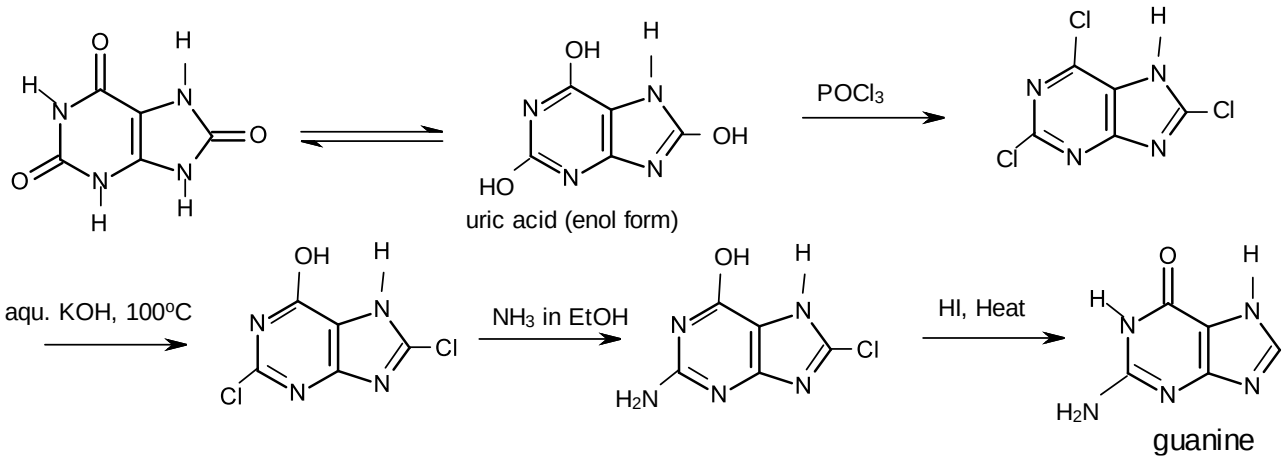
Uric acid is treated with methyl iodide in alkaline solution to give 1,3,7-trimethyl uric acid. This on heating with $POCl_3$ to form chlorocaffeine which on reduction with HI yields caffeine.



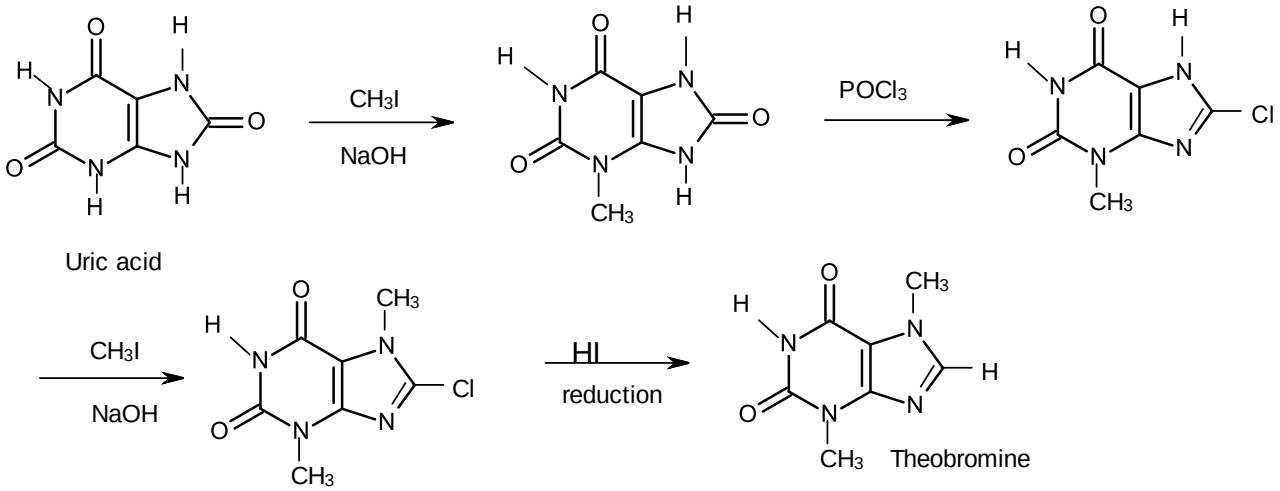
Conversion of uric acid to purine.



Synthesis of Guanine:



Synthesis of Theobromine:



3. ALKALOIDS

Alkaloids (meaning alkali like) are an important class of basic organic compounds that occur in higher plants. They contain one or more nitrogen containing heterocylic rings as an integral part of their structure.

Alkaloids are generally defined as the compounds of plant origin having one or more basic nitrogen containing heterocyclic rings and can cause marked physiological activity in human beings.

Classification: Alkaloids are divided into groups or classes depending on the prominent heterocylic ring system present in the molecule. The various groups of alkaloids are

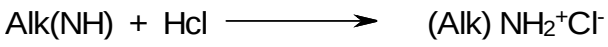
1. Pyridine alkaloids: prominent heterocyclic ring; pyridine , e.g. nicotine
2. Piperidine alkaloids: Prominent heterocylic ring – piperidine, e.g. coniine, piperine

3. Quinoline alkaloids: containing quinoline as the prominent heterocylic ring. E.g. quinine.
4. Isoquinoline alkaloids: containing isoquinoline ring e.g. papaverine, morphine
5. Indole alkaloids : heterocylic ring is indole e.g. gramine, strychnine

GENERAL PROPERTIES:

1. Alkaloids are usually colourless, crystalline, non-volatile solids while a few of them (e.g. coniine , nicotine) are liquids.
2. In general they are insoluble in water but liquid alkaloids are soluble. They dissolve readily in organic solvents such as ethanol, ether or chloroform.
3. They are optically active, the majority of them being laevorotatory.
4. They are basic with a bitter taste & dissolve in mineral acids to form salts

5. Most of the alkaloids are tertiary bases while a few are secondary bases. They form well defined crystalline salts with mineral acids.



PHYSIOLOGICAL ACTIVITY:

The Physiological activity of alkaloids are quite diverse.

Morphine is a powerful narcotic used for the relief of pain, though its addictive properties limit its usefulness.

Codeine, the methyl ether derivative of morphine found in the opium poppy, is an excellent analgesic that is relatively nonaddictive.

Certain alkaloids act as cardiac or respiratory stimulants. Quinidine, which is obtained from plants of the genus *Cinchona,* is used to treat arrhythmias, or irregular rhythms of the heartbeat.

Ergonovine is used to reduce uterine hemorrhage after childbirth,

Ephedrine is used to relieve the discomfort of common colds, sinusitis, hay fever, and bronchial asthma.

Many alkaloids possess local anesthetic properties, though clinically they are seldom used for this purpose.
Cocaine (from *Erythroxyloncoca*) is a very potent local anesthetic.
Quinine (from *Cinchona*species) is a powerful antimalarial agent that was formerly the drug of choice for treating that disease, though it has been largely replaced by less toxic and more effective synthetic drugs.

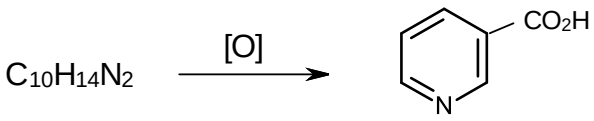
Some alkaloids are illicit drugs and poisons. These include the hallucinogenic drugs mescaline (from *Anhalonium* species) and psilocybin (from *Psilocybe mexicana*).

Synthetic derivatives of the alkaloids morphine and lysergic acid (from *C. purpurea*) produce heroin and LSD, respectively. The alkaloid coniine is the active component of the poison hemlock(*Conium maculatum*).

Strychnine (from*Strychnos* species) is another powerful poison

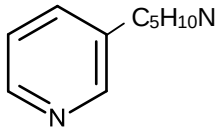
ISOLATION OF ALKALOIDS:
Alkaloids are usually found in the seeds, roots, leaves or bark of the certain plants.
The plant material dried, finely powdered and extracted with boiling methanol. The cellulosic material present in extract is separated by filtration. The filtrate is evaporated to give the crude plant extract. This is then dissolved in dil. Acid and extracted with ether. NaOH is then added to the acid solution and the resulting basic solution is also extracted with ether. Evaporation of ether extracts gives a solid mixture of crude alkaloids. It is then subjected to fractional crystallization or chromatography to separate individual alkaloids.

CONSTITUTION OR STRUCTURAL ELUCIDATION OF NICOTINE:
Nicotine is found in tobacco plant.
1. The molecular weight determination and elemental analysis leads to the molecular formula of $C_{10}H_{14}N_2$
2. Presence of two 3° nitrogen atoms: when treated with methyl iodide, it forms dimethiodide showing that both the N atoms of nicotine are tertiary in character.
3. Presence of a pyridine ring: on oxidation with chromic acid or nitric acid nicotine gives nicotinic acid (pyridine-3-carboxylic acid)



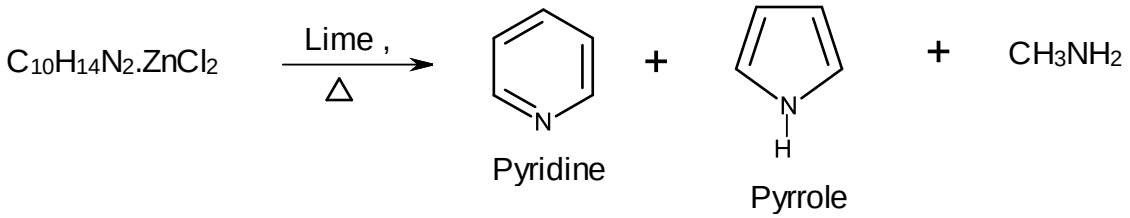
Nicotinic acid

This shows that this alkaloid contains a pyridine ring with a side chain. Therefore the formula of



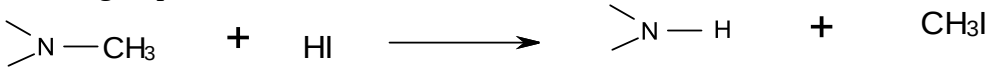
nicotine may be written as

4. **Nature and position of side chain**:
Nicotine forms an addition compound with zinc chloride $C_{10}H_{14}N_2.ZnCl_2$ which when heated with lime yields pyridine, pyrrole and methyl amine

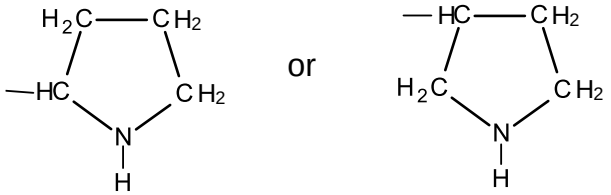


This suggests that the side chain $C_5H_{10}N$ is a pyrrole derivative

5. When heated with conc. hydroiodic acid at 200-300° C nicotine yields CH_3I.. This shows that CH_3 group in the side chain is attached to N atom.

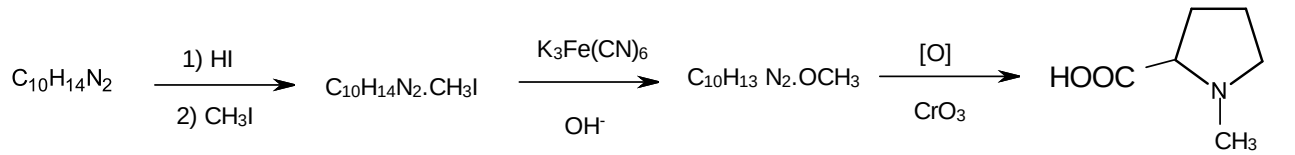


Therefore the side chain could be N – methyl pyrrolidine.



However the point of attachment of the side chain to the pyridine nucleus could be C_2 or C_3 as shown above.

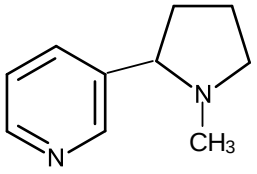
6. Nicotine hydroiodide when treated with methyl iodide forms nicotine isomethiodide which on oxidation with potassium ferricyanide yields nicotone. Nicotone on oxidation with CrO_3 produces hygrinic acid.



Hygrinic acid

The formation of hygrinic acid proves that the side chain N-methyl pyrrolidine is attached to the pyridine nucleus through C_2

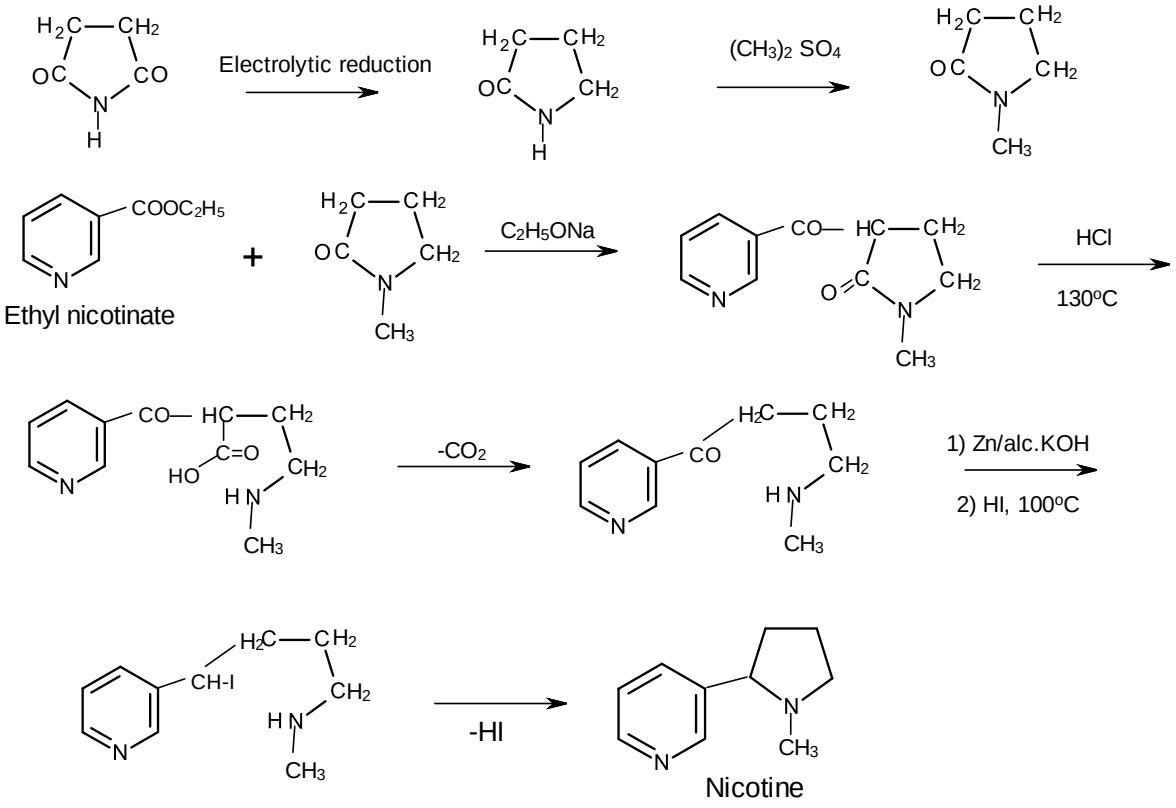
7. By considering above facts the structural formula of nicotine may be constructed as



Nicotine

SYNTHESIS

Finally the structure of nicotine is confirmed by synthesis.



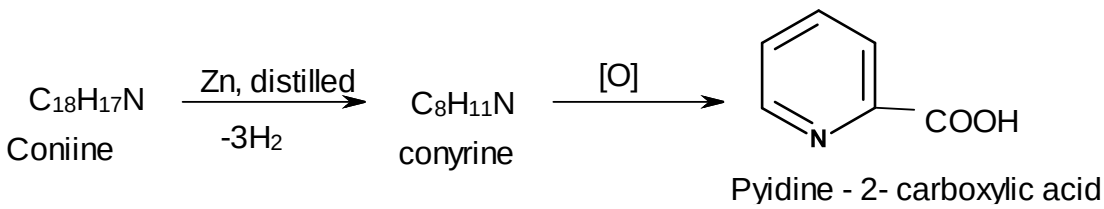
Structural Elucidation & synthesis of coniine

CONIINE

It is the principal toxic substance present in the hemlock herb (conium maculatum)

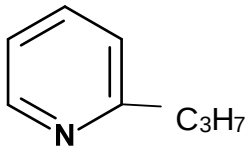
Structural elucidation: 1. The molecular formulal of coniine is $C_8H_{17}N$

2. when distilled with zinc dust coniine loses six hydrogens giving a new base conyrine $C_8H_{11}N$. The conyrine on oxidation with $KMnO_4$ yeilds pyridine – 2- carboxylic acid (α – picolinic acid)

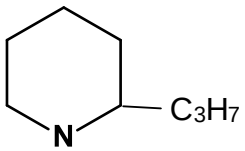


This shows that conyrine is made up of a pyridine ring with a side chain at 2-position.

And also conyrine is obtained from coniine by loss of six hydrogen atoms, the structure of conyrine & coniine may be written as



conyirine

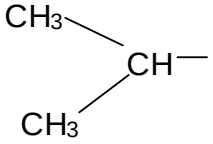


coniine

Now the side chain (C_3H_7) in coniine can be represented in two ways.

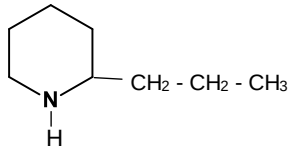
$CH_3 - CH_2 - CH_2 -$

n-propyl

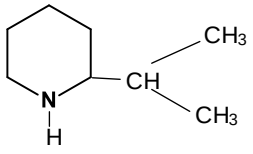


isopropyl

Therefore two possible structures of coniine are

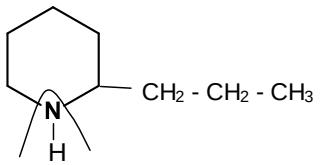


I



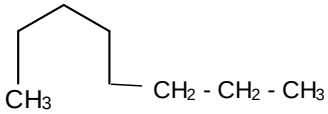
II

On heating with hydroiodic acid & phosphorus , coniine is reduced to give n- octane. Formation of n-octane is possible only when the side chain is n- propyl as in structure I . Therefore the structure of coniine is structure I

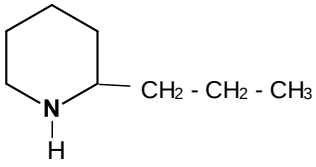


I

HI/P



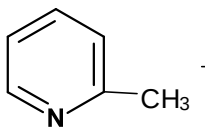
n- octane



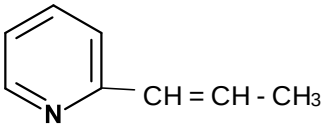
coniine

Synthesis : Landenburg's synthesis

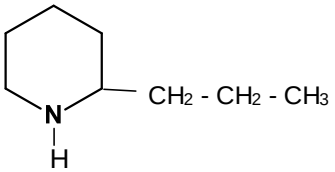
α – Picoline (2- Methyl pyridine) is condensed with acetaldehyde at 250° C to form 2 – propenyl pyridine which on reduction with sodium and ethanol yields coniine



CH_3CHO
250°C

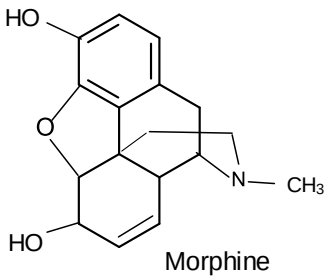


Na/C_2H_5OH

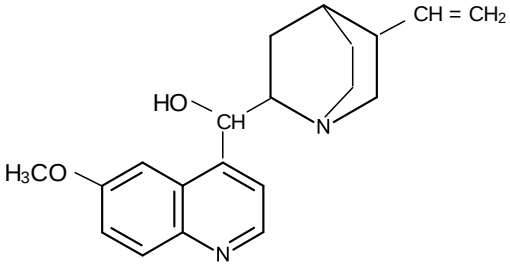


(±) coniine

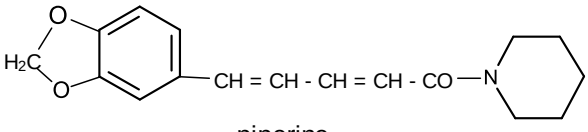
Structural formulae of quinine, atrophine, piperine, cocaine and morphine.



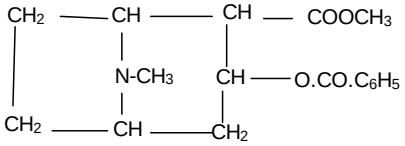
Morphine



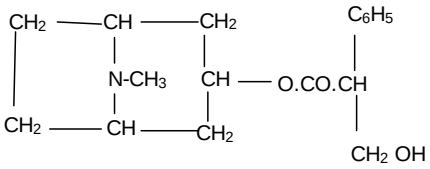
Quinine



piperine



Cocaine



Atropine

Quinine is antipyretics, antimalarial

Cocaine acts as local anaesthetic

Morphine relieves pain (analgesic)
atrophine dilates pupil of the eye. , anticholinergic,
Nicotine is stimulant

VITAMINS AND HORMONES

Organic compounds which are necessary for the life, growth and health of animals in addition to oxygen, water, proteins, carbohydrates etc. are known as 'accessory dietary factors' called vitamins.

Vitamins are only necessary in very small amounts. Vitamins cannot be produced by the body and hence must be supplied through food. The lack of vitamins in the diet causes deficiency diseases such as beriberi, scurvy, rickets etc. Vitamins do no serve as source of energy.

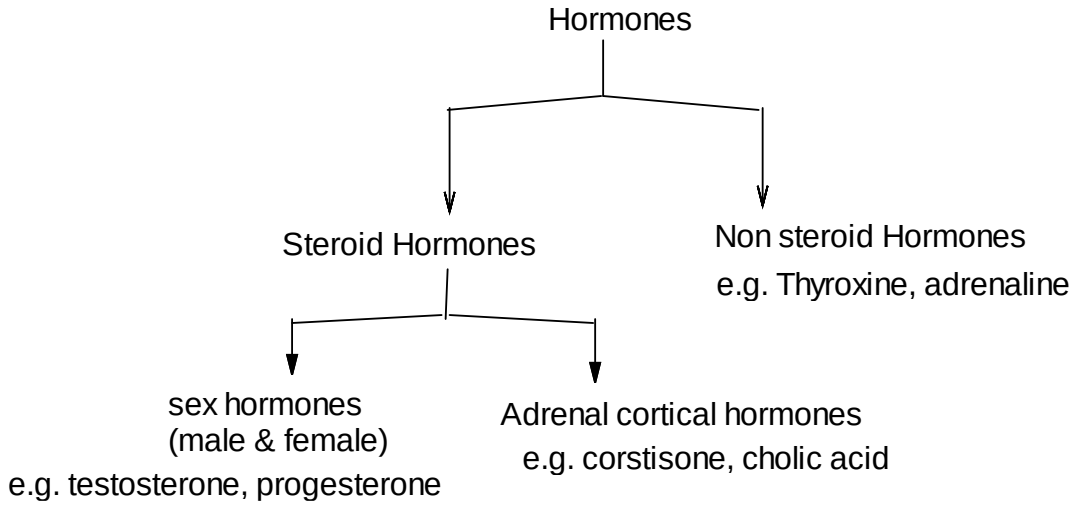
Some important vitamins and their sources are given below.

Vitamin A (A_1) and A_2 –fish lever oil.
Vitamin B_1, B_2 and B_6 – rice polishings, egg, yeast
Vitamin C – green vegetables, cabbages, citrus fruits.
Vitamin D_1, D_2 and D_3 – cod liver oil.
Vitamin E – wheat germ oil.
Classification:
Vitamins are classified into two types –
1) Fat soluble vitamins (vitamins A, D, E & K)
2) Water soluble vitamins (vit B & C)

Hormones: Hormones are chemical compounds secreted by ductless glands such as thyroid, adrenals & pancreas etc. they catalyze and control diverse metabolic process and hence the name hormones. They are highly specific in their physiological action. Eg. Adrenaline, thyroxine etc.

Classification: Hormones are classified into two types i) Steroid hormones ii) Non-steroid hormones.

Steroid hormones are again divided into 2 types i) Sex hormones (male & female) ii) adrenal cortical hormones.

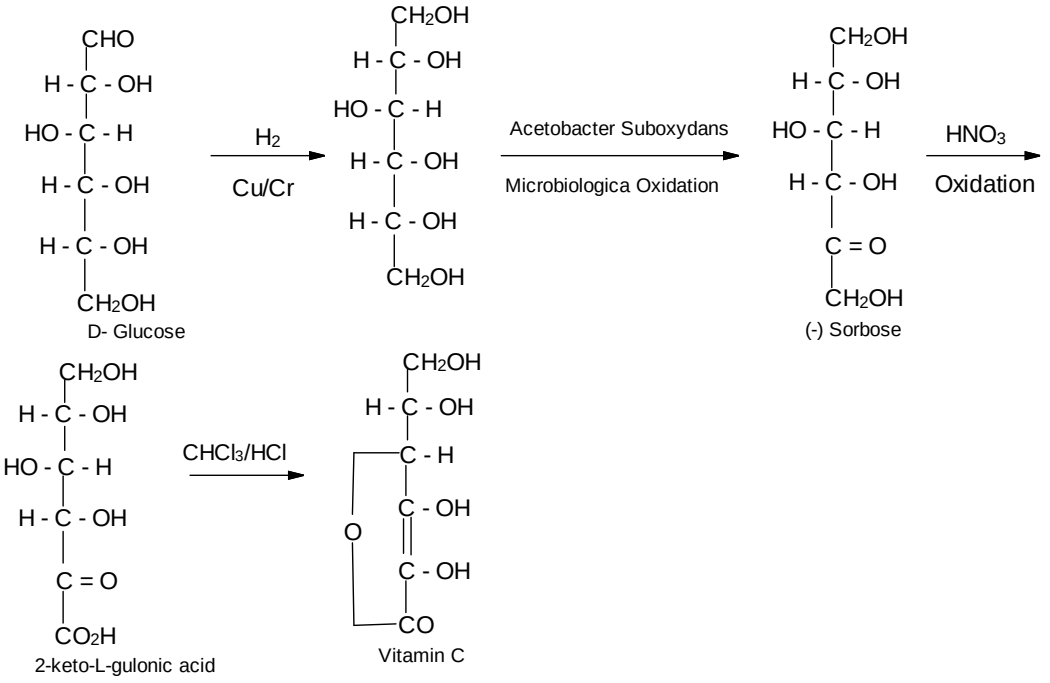


+

Vitamin C (L- ascorbic acid)

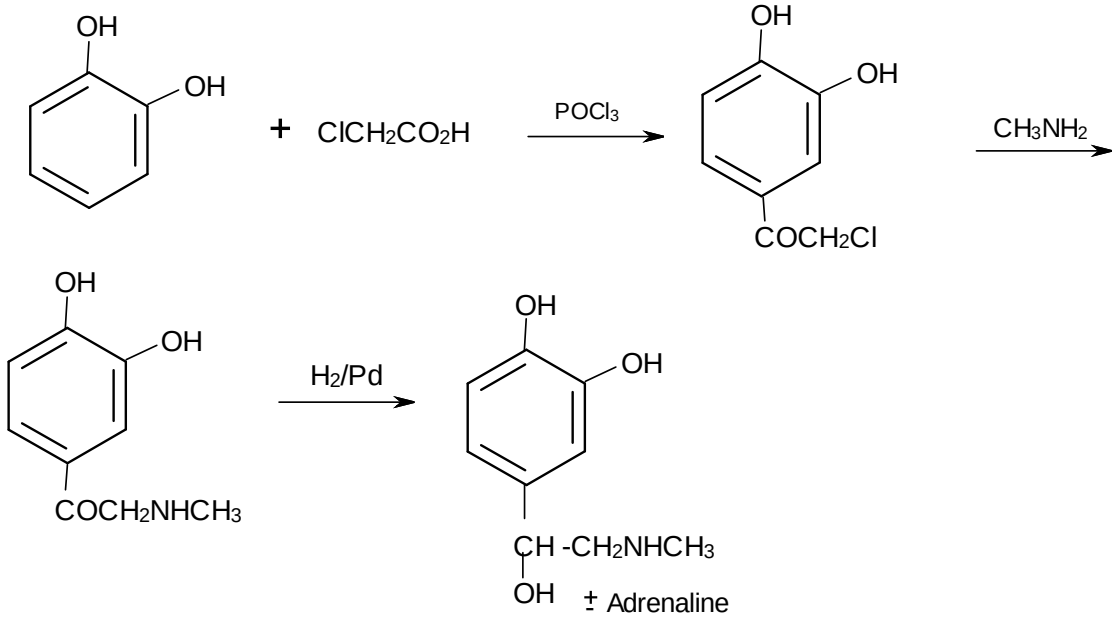
Vitamin C is found in green vegetables (beans, tomato, cabbage etc.) and citrus fruits like lemon, oranges etc. deficiency of vit C caused the disease called scurvy, brittleness of bones and decrease of resistance of disease.

Synthesis: Vitamin C is synthesized form D-glucose as follows.

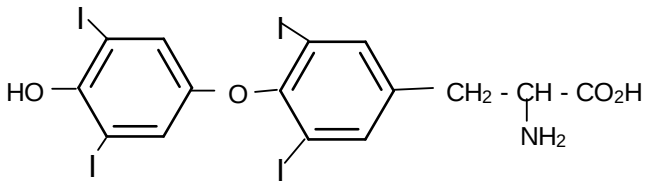


Synthesis of adrenaline: (Apinephrine)

It is the hormone secreted by adrenal medulla. Adrenaline is used in the treatment of asthma. It increases blood pressure when injected to the blood stream and is used locally to stop hemorrhage



Structural formula of Thyroxine



Thyroxine

Vitamin A (A_1) ,Retinol- structural elucidation

1. Elemental analysis and molecular weight determination of vit A shows that its molecular formula to be $C_{20}H_{30}O$

2. On catalytic hydrogenation, Vit A is converted into perhydro vitamin A, $C_{20}H_{40}O$. it shows the presence of five double bonds in vit A.

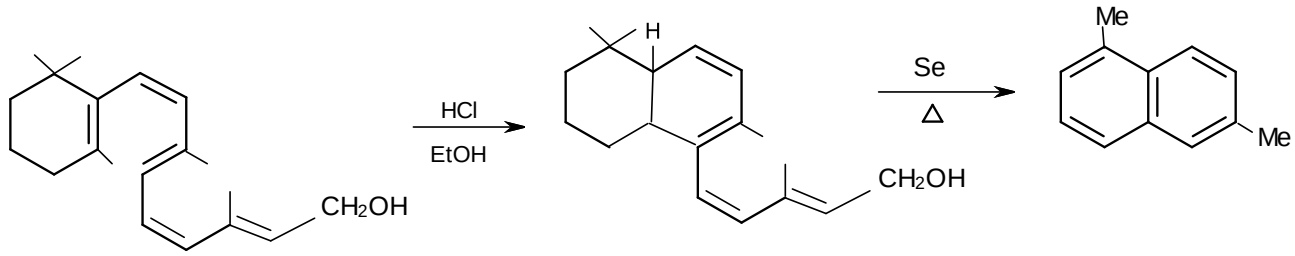
3. vit A forms an ester with p-nitro benzoic acid. It means that vit A contains an alcoholic group.

4. ozonolysis of vit A produces one molecule of geronic acid per molecule of vit A. this indicates the presence of one β- ionone nucleus.

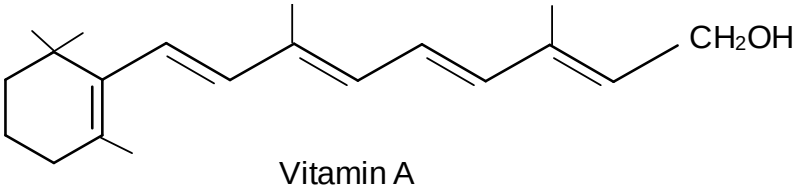
5. oxidation of vit A with $KMnO_4$ produces acetic acid . This shows the presence of some – $C(CH_3)$= groups in the chain.

6. We know that β-carotene is converted into vit A in the intestinal mucosa. This suggests that vit A is half of β-carotene structure.

7. on heating vit A in an ethanolic solution of hydrogen chloride, compound I is obtained which on dehydrogenation with selenium forms 1: 6 – dimethylnaphthalene. The following structure of vit A explains these reactions.

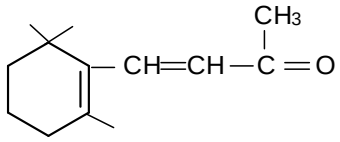


8. Perhydro vitamin A has been synthesized from β-ionone which is identical with that obtained by reducing vit A. This lends support to the above structure of vit A.
From the above evidences structure of vit A is



Vitamin A

Synthesis of vitamin A



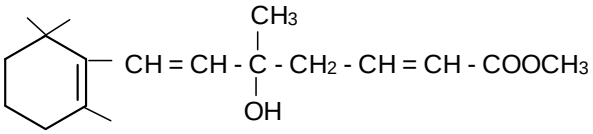
β - ionone

+ $BrCH_2CH = CH - COOCH_3$

Methyl γ -bromocrotonate

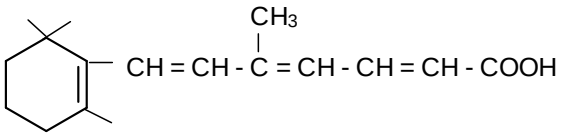
Zn

Reformatsky reaction



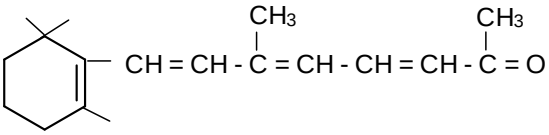
1) anhy. oxalic acid

2) KOH



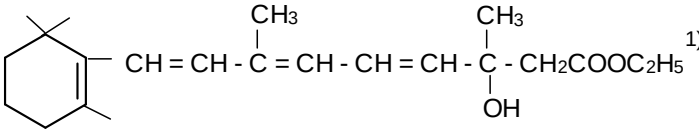
1) $SOCl_2$

2) CH_3Li



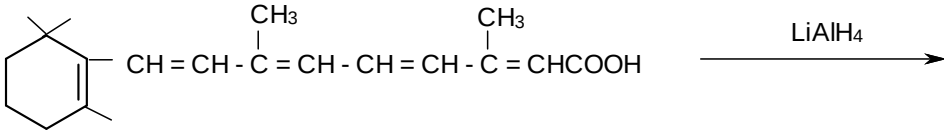
$BrCH_2COOC_2H_5$ / Zn

Reformatsky reaction

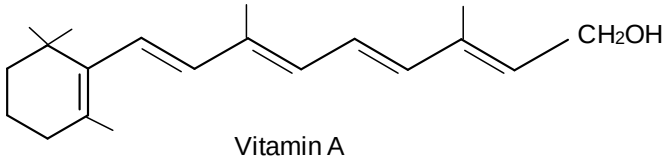


1) anhydrous oxalic acid

2) KOH



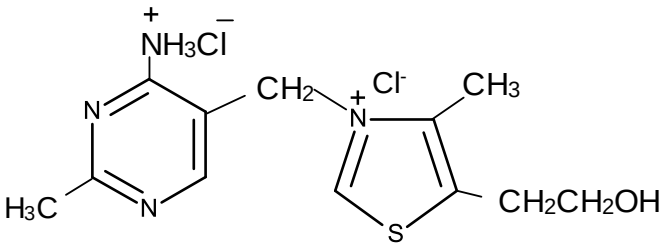
$LiAlH_4$



Vitamin A

Vitamin B_1 (thiamine) (aneurin)

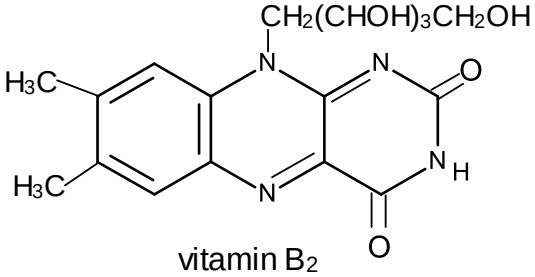
Thiamine is one of the member of the water soluble vit B complex. Vitamin B_1 occurs in rice polishings, yeast, egg etc. Deficiency of vit B_1 causes beriberi in man.



Vitamin B_1

Vitamin B_2, Riboflavin (lacto flavin)

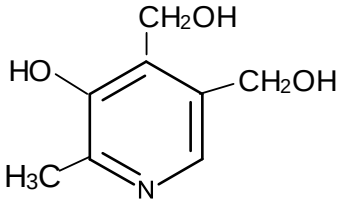
Riboflavin is the water soluble, thermostable vitamin which occurs in the vitamin B complex. It is necessary for growth and health. It occurs widely in yeast, green vegetables, milk, meat etc. Since it was first isolated from milk, it is also known as lactoflavin.



vitamin B_2

Vitamin B_6 or pyridoxine:

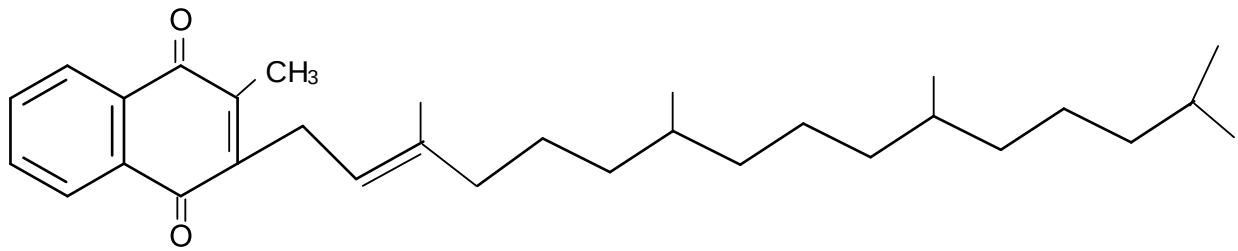
It is obtained from rice bran and yeast. It cures dermatitis is rats.



Vitamin B_6

Vitamin K (K_1)

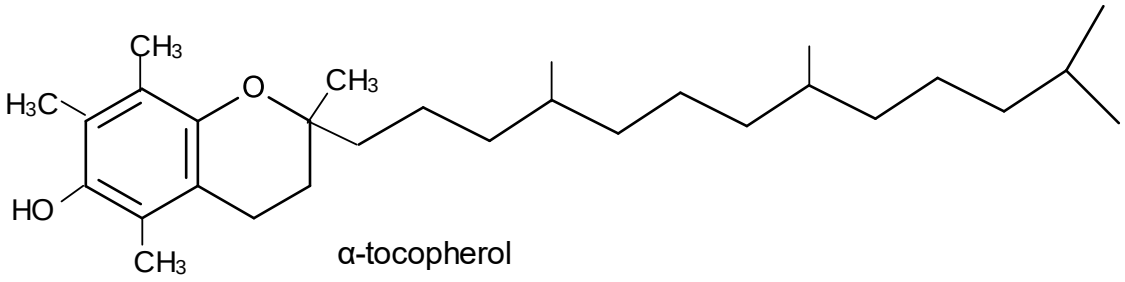
The best source of vitamin K is alfalfa, cabbage, spinach and carrot tops. Both vitamin K_1 and K_2 are anti hemorrhagic vitamins. They are connected with the enzymes involved in blood clotting; a deficiency of them will lengthen the time of blood clotting.



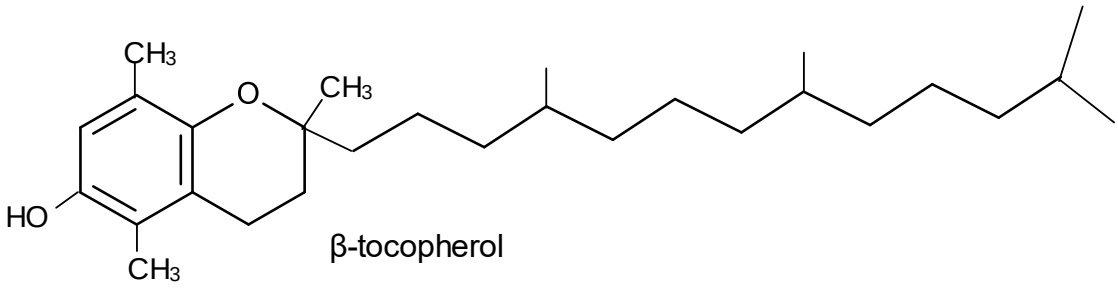
Vitamin K

Vitamin E or Tocopherols:

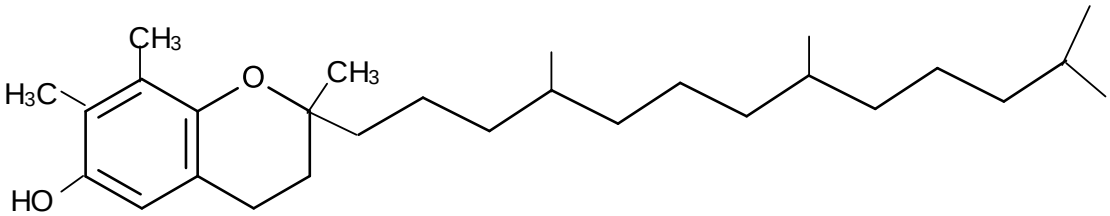
The term vit E refers to a group of closely related compounds which occur naturally. They are, to different degrees, anti- sterility factors. Eight compounds of these are collectively called tocopherols. The most important are α, β, γ - tocopherols. The main source of α & β- tocopherols is wheat germ oil. γ –compound is obtained from cotton seed oil.



α-tocopherol



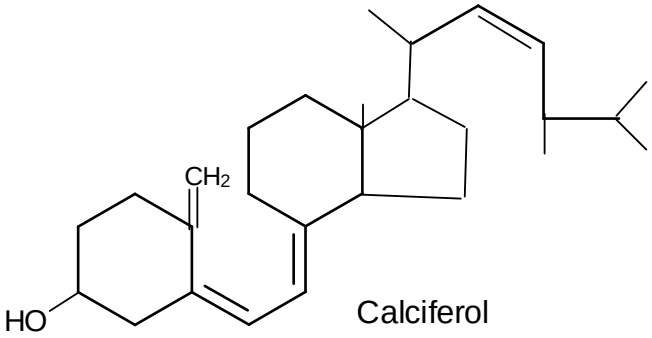
β-tocopherol



γ-tocopherol

Calciferol or vit D or D_1 or D_2 (also known as ergo calciferol)

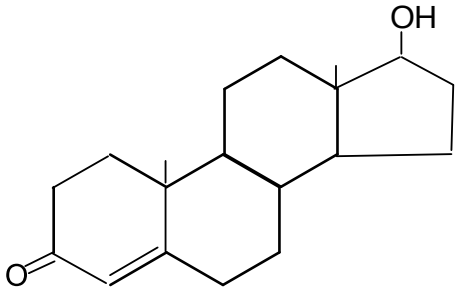
This vitamin is the anti rachitic vitamin. It is essential for bone formation. its function being the control of calcium and phosphorous metabolism.



Calciferol

Structure & importance of testosterone:

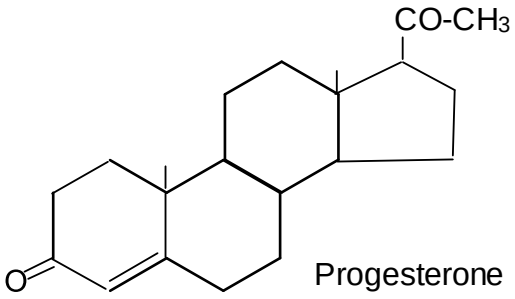
It is secreted mainly by the testicles and is responsible for the various male sexual characters. It is the major male sex hormone and the most potent one. It stimulates the normal development of male reproductive organs and also the development of the secondary male sex characteristics.



Testosterone

Structure and importance of progesterone

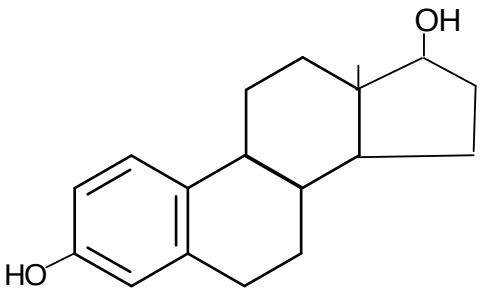
It is the one of the most important female sex hormones. It acts mainly in the uterus to permit implantation and gestation.



Progesterone

Structure and importance of Estradiol:

Estradiol is responsible for the maintenance of the female reproductive organs. It is synthesisezed mainly in the ovary and placenta.



estradiol

DRUGS

Compounds which exert various physiological effects of therapeutic value are collectively called drugs or chemotherapeutic agents. .

Chemotheraphy is the use of chemicals(drugs) in the treatment of infectious diseases, so as to destroy the invading microorganisms without causing considerable damage to the host tissues.

TYPES OF DRUGS:

1. Antipyretics: Antipyretics are substances that lower the body temperature. E.g. Aspirin, Paracetamol, Analgin, Phenacetin etc.

2. Analgesics: Analgesics are substances that are used to relieve pain. These are of two types i) Narcotics
ii) Non- narcotics.

i) Narcotics: Narcotics induce sleep and unconsciousness in higher doses. They cause addiction. e.g. they are mainly products of opium such as morphine, heroin etc.

ii) Non- narcotics: this group of drugs also have antipyretic properties. They do not induce sleep. E.g. aspirin, analgin.

3. Anaesthetics: these produce loss of sensation. These are of two types:
i) Local anaesthetics: which produce loss of sensation on a small area where the drug is applied. They are used for small surgical operations. E.g. Xylocaine, ethyl chloride, procaine etc.
ii) General anaesthetics: they produce unconsciousness and used in major surgical operations.
e.g. chloroform, nitrous oxide, cyclopropane etc.

4. Sedatives: these are the drugs used for patients who are violent or mentally agitated. Eg. diazepam, meprobamats.

5. Antiseptics: antiseptics are applied to living tissues like wounds, cuts etc. Antiseptics may kill the bacteria or prevent its multiplication. E.g. Dettol, Savlon, boric acid. Etc.

6. Antibacterials : These are called as sulpha drugs and were the first effective chemotherapeutic agents to have been employed for the prevention and cure of bacterial infections.
e.g. sulphanilamide, sulphapyridine, sulphathiazole etc.

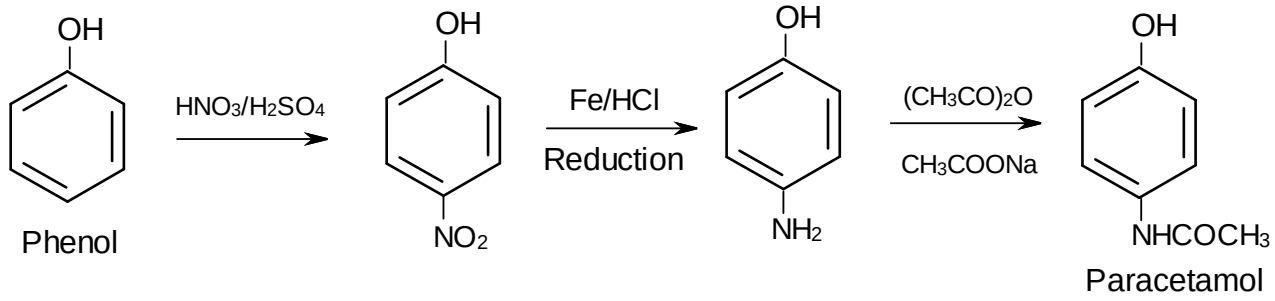
7. Antibiotics: An anti biotic is a substance produced by microorganisms that is capable of inhibiting or destroying the growth of other micro-organisms.

In 1929, Fleming discovered a mould of the penicillium species which inhibited the growth of certain bacteria. This observation was investigated later by a number of workers and isolated active penicillin. The antibiotics popular in current medical practice include Penicillin, streptomycin, chloramphenicol, tetracyclines, erythromycin
Cholamphenicol (chloromycetin) is the first antibiotic produced commercially by synthetic process.

8. Antimalarials: The drugs used for suppression and treatment of disease malaria are called antimalarials. The cinchona alkaloid quinine was originally the only drug known to be effective against malaria. Quinine has now been replaced by synthetic antimalarials. Few of them are chloroquine, primaquine, proguanil, plasmoquin etc.

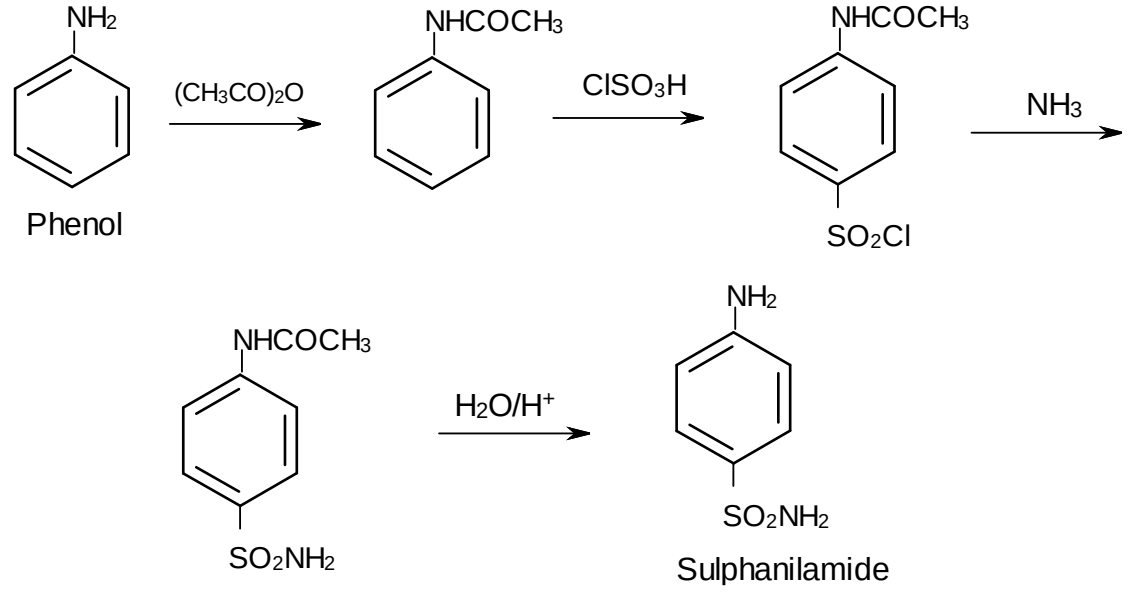
Synthesis of Paracetamol: (p-acetamido phenol or N-acetyl p-aminophenol)

It can be synthesized from phenol as follows.

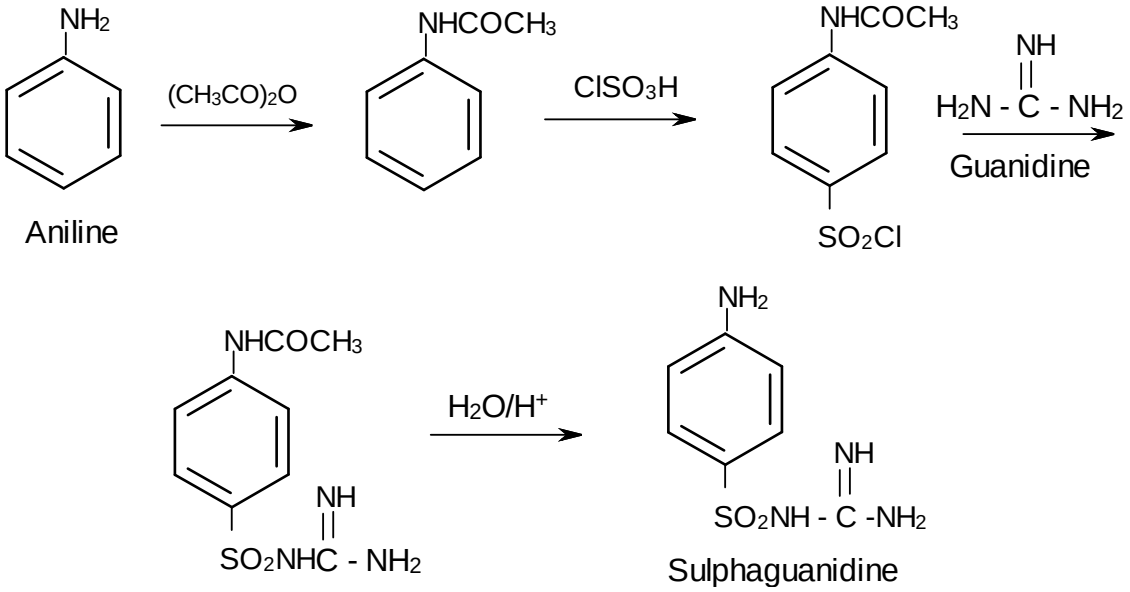


Uses: it has analgesic and antipyretic activities. It is being used as such under various trade names like crocin, metacin etc.

Synthesis of sulphanilamide: sulphanilamide is synthesized from aniline by the following steps.

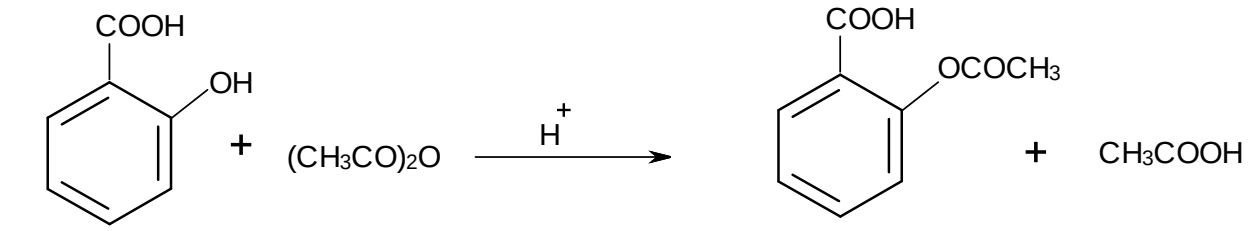


Synthesis of sulphaguanidine: sulphaguanidine is synthesized from acetanilide as follows.



Synthesis of aspirin: (Drugs chapter)

It is prepared by heating salicylic acid with acetyl chloride or acetic anhydride in the presence of phosphoric acid.



salicylic acid

Acetyl salicylic acid
(Aspirin)

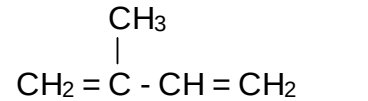
Uses: It is used to reduce fewer. It is also used as a pain killer. It is sold under the trade name aspirin.

TERPENES

Terpenes occur widely in the leaves and fruits of certain plants such as conifers, citrus and eucalypts. When the plant source is steam distilled, the oily substance that obtained is called essential oil. It is a mixture of various terpenes.

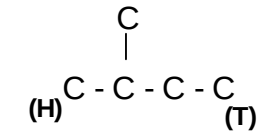
Definition: Terpenes are essential oils responsible for odours and flavour of plants. They are hydrocarbons having the general formula $(C_5H_8)_n$ or their oxygenated derivatives.

Isoprene rule: states that the molecules of all terpenes are constructed of two or more isoprene units usually joined in a head to tail fashion. Isoprene is 2-Methyl-1,3-butadiene

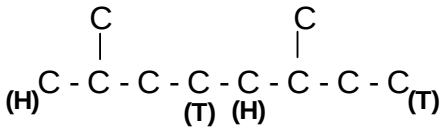


2- Methyl - 1,3 - butadiene

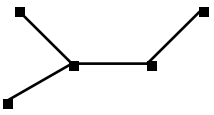
Terpenes are obtained by linking together various isoprene units through carbon atom no. 1 and carbon atom no.4 (head to tail linkage)



isoprene unit



Two isoprene units linked head to tail



Isoprene skeleton

Classification: Terpenes are classified according to the number of isoprene units in the molecule. The simplest terpenes have 2 isoprene units (ten carbon atoms) and are called monoterpenes.

Other classes are

Monoterpenes – two isoprene units, C_{10}

Sesqiterpenes – three isoprene units, C_{15}

diterpenes – four isoprene units, C_{20}

triterpenes – six isoprene units, C_{30}
tetraterpenes – eight isoprene units, C_{40}
each class of terpenes are further subdivided according to the number of rings in the molecule. They are i) Acyclic – no rings (straight chain)
ii) monocyclic – one ring
iii) bicyclic - two rings
iv) tricyclic – three rings.

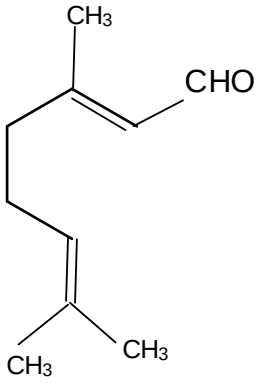
ISOLATION OF TERPENES

1) Steam distillation: The plant material is macerated and steam distilled. The essential oil distills over. The aqueous steam distillate is saturated with salt and extracted with light petroleum or benzene. The solvent extract is dried and the solvent is removed by evaporation under reduced pressure to give the essential oil.

2) Direct solvent extraction:
If a particular terpene is decomposed under the conditions of steam distillation, the plant material is directly extracted with light petroleum or ether at room temperature. The extract is filtered and the solvent is removed by evaporation under vacuum to recover the essential oil. The essential oils obtained as above are usually mixture of many terpenes. These are separated by fractional distillation or vapour phase chromatography.

Citral

It is the most important acyclic (open chain) monoterpenoid and has the structural formula as follows.

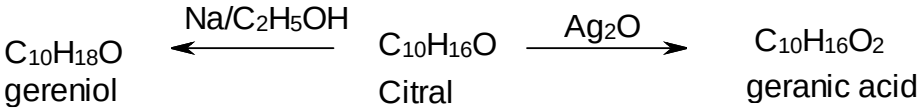


Citral

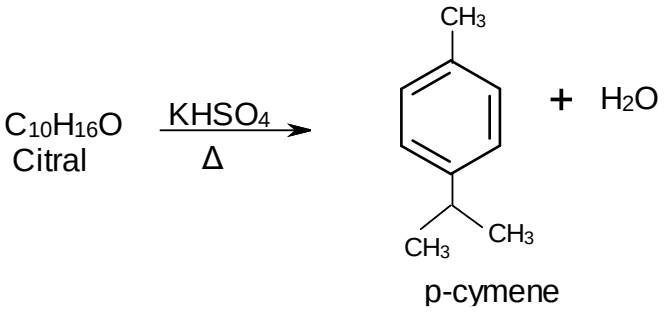
Citral occurs to the extent of 80% in lemon grass oil. It is also present in the oil of citrus fruits, (Citron, orange, lemon)

<u>Elucidation of structure of citral</u>:
1. Elemental analysis and molecular weight determination shows that molecular formula of citral is $C_{10}H_{16}O$
2. Citral reacts with bromine (2 molecules) to form tetrabromide derivative. This indicates the presence of 2 carbon – carbon double bonds in the citral molecule.

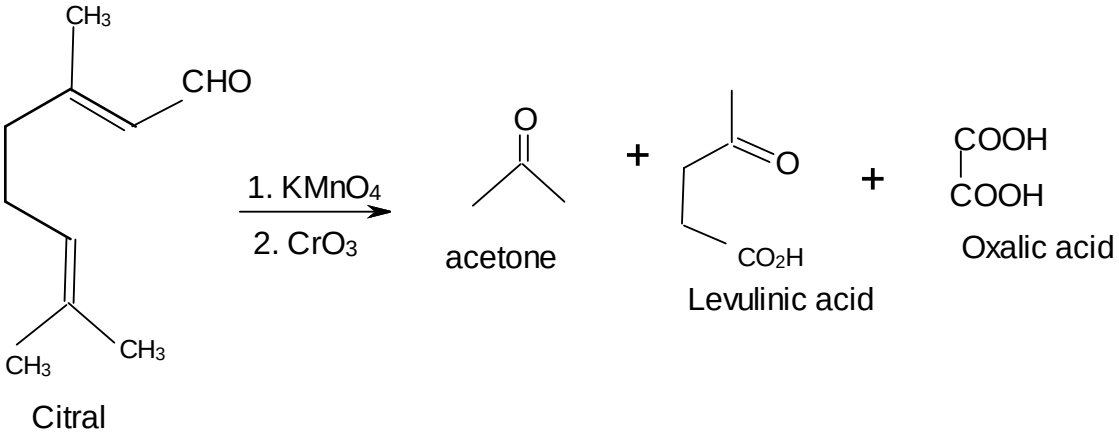
3. Citral reacts with hydroxylamine to form an oxime. It also adds sodium bisulphite, indicating the presence of a carbonyl group in citral.
4. Citral on oxidation with silver oxide gives geranic acid and on reduction with Na/alcohol gives gereniol, a primary alcohol. These reactions indicates that the carbonyl group present in citral is in the form aldehyde group.



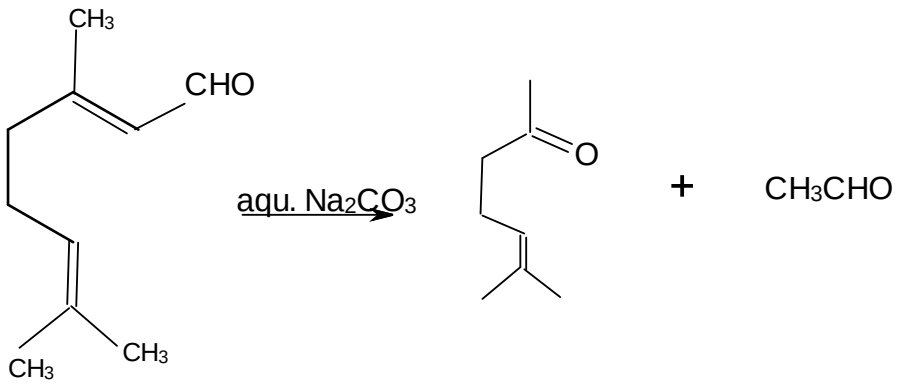
5. when citral is heated with $KHSO_4$, it is converted to p-cymene. The reaction indicates the relative position of methyl and isopropyl groups in citral.



6. Position of C=C double bonds: the oxidation of citral with alkaline permanganate followed by chromic acid gives acetone, levulinic acid and oxalic acid. This is accountable only if the position of the double bonds is as shown in the formula below

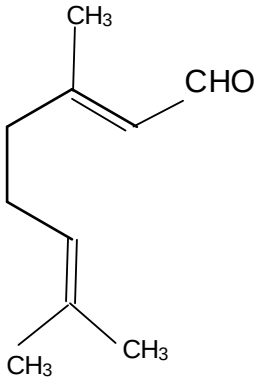


7. The position of double bonds is further supported by the decomposition of citral with aq. Na_2CO_3 to give methyl heptenone and acetaldehyde.



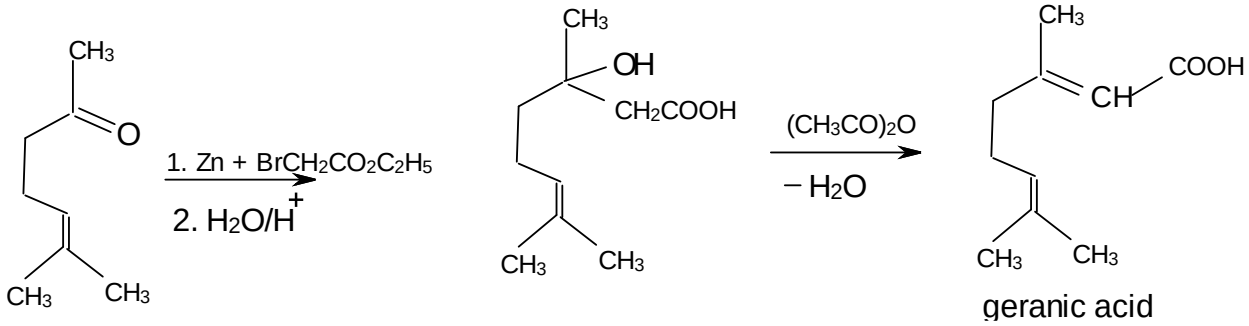
Citral

On the basis of above experimental evidence the following structure is assigned to citral.

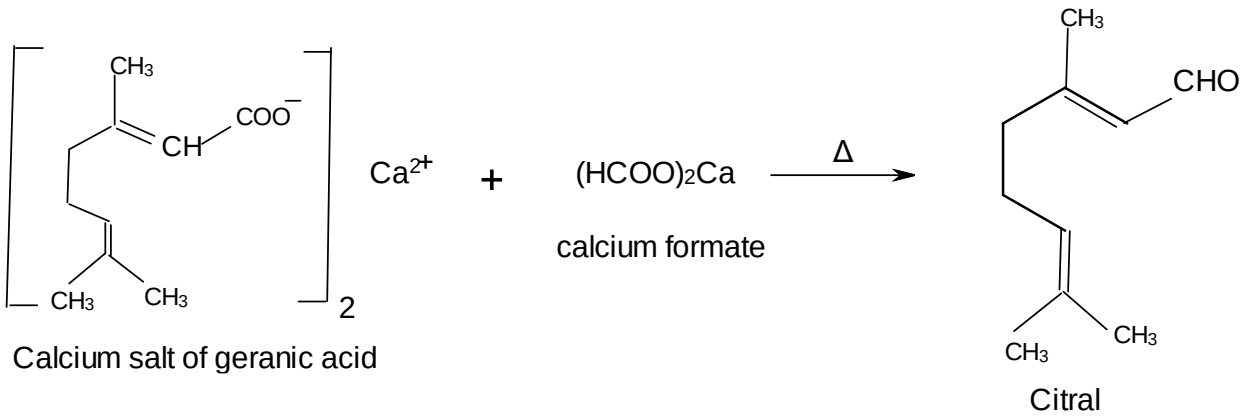


Citral

Synthesis: the structure of citral was confirmed by the following synthesis from methyl heptenone



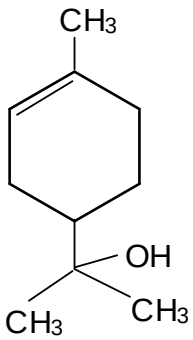
geranic acid



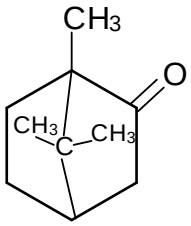
Calcium salt of geranic acid

Citral

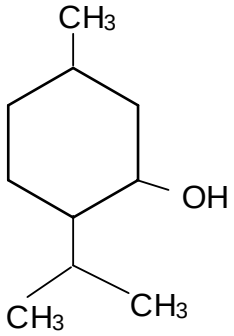
Structural formula of α – terpeneol, camphor & Menthol



α - terpeneol



camphor



Menthol

Structural Elucidation & synthesis of Limonene

Limonene

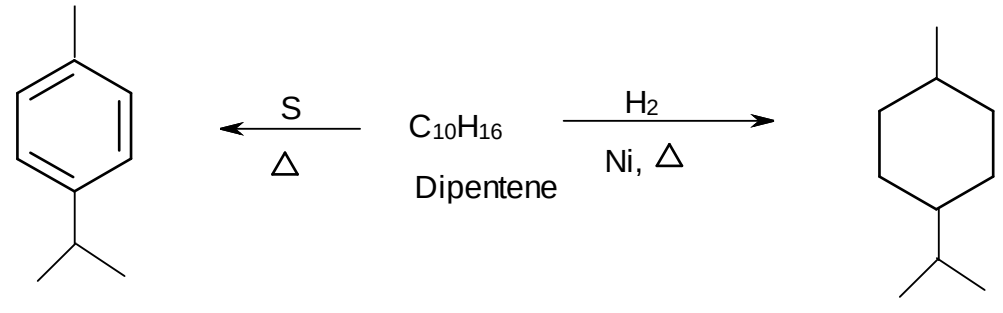
Limonene is an important monoterpene. It is optically active hydrocarbon. (+) Limonene occurs in the oils of orange, lemon and lemon grass. (-) Limonene is found in pine needles and in peppermint oil.

(±) Limonene also known as dipentene occurs in turpentine oil.

Structure: 1. M. F. of limonene is $C_{10}H_{16}$

2. Dipentene (Limonene) reacts with 2 molecules of bromine to form tetrabromide. It also reacts with HBr(2 molecules) to give the corresponding dihydrobromide. These reactions indicates the presence of 2 carbon carbon double bonds in dipentene.

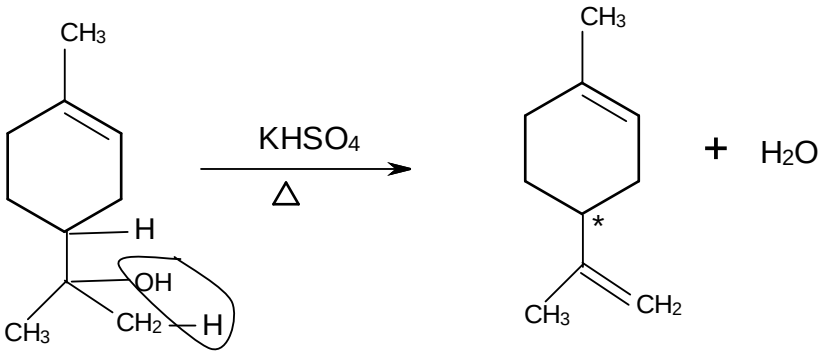
3. Dipentene undergoes dehydrogenation when heated with sulphur to yield p-cymene. It also catalytic hydrogenation to give menthane. These reactions are giving the idea of nature of carbon frame in limonene



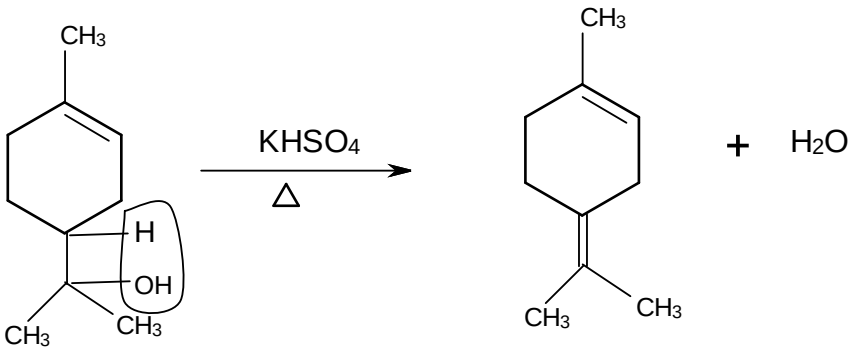
p- cymene p-Menthane

Dipentene can be obtained by dehydration of α – terpeneol with potassium bisulphate or oxalic acid.

Dehydration of α – terpeneol can takes place in two ways which are given below.



Structure 'A'

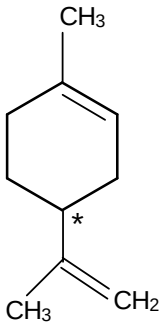


Structure 'B'

Structure A is optically active because there is on asymmetric carbon atom, where as structure B is optically inactive.

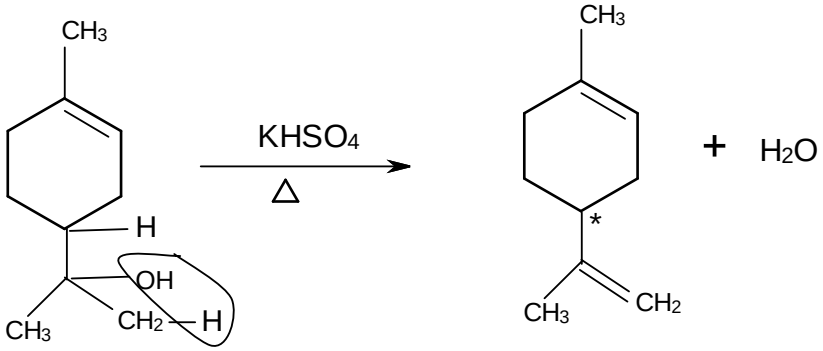
Therefore, Structure A is structure of limonene (dipentene)

Therefore, structure of dipentene is

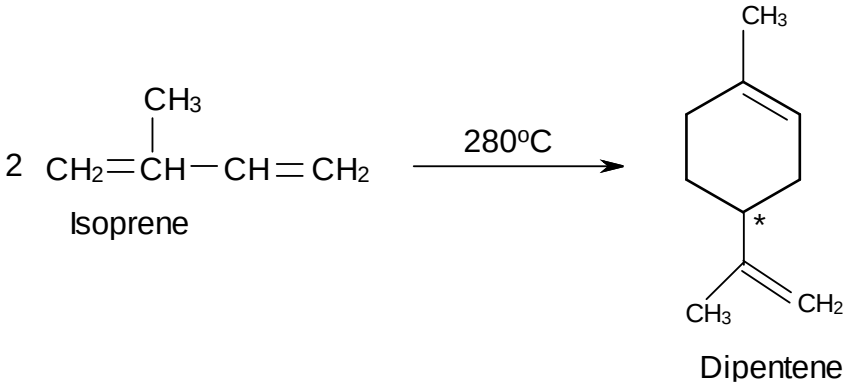


Synthesis: Dipentene may be obtained

1. by heating α – terpeneol with potassium bisulphate



2. By heating isoprene in a sealed tube at 280°C



UV SPECTROSCOPY

When a beam of electromagnetic radiations (U.V. or visible) is passed through a compound, certain radiations or wave lengths are absorbed by the substance and causes electronic excitation i.e., electrons are excited to higher levels which results in UV spectrum.

It is important to note that the molecule absorbs only that radiation which possess the appropriate energy to bring about permitted transition to higher electronic, vibrational, or rotational levels. This all depends upon the structure of the molecule. The radiations absorbed is characteristic of a substance.

The range of UV spectroscopy is 200 – 400 nm.

Electronic transitions: when a molecule absorbs radiations the electrons are excited to higher levels. The electron involved could be σ electrons (occupying σ molecular orbitals) or π

electrons (occupying a π molecular orbitals) or n electron (non-bonding). The following electronic transitions are possible.

i) $\sigma \rightarrow \sigma^*$ transition: Transition in which a σ electron is excited to σ^* orbital is called $\sigma \rightarrow \sigma^*$ transition. A high amount is required for this transition. This energy is supplied by UV radiation fo wavelength less than 200 nm.

ii) $n \rightarrow \sigma^*$ transition: this transition corresponding to the excitation of an nonbonding electron to σ^* molecular orbital. These transitions are associated much lower energy than $\sigma \rightarrow \sigma^*$ transitions.

iii) **$n \rightarrow \pi^*$ transition**: this transition corresponding to the excitation of an nonbonding electrons to π^* molecular orbitals. compounds having double or triple bonds such C= O or C = S and N = S display this transitions. Small amount of energies are required for this transitions which takes place in the ordinary UV range.

i) **$\pi \rightarrow \pi^*$ transitions**: these are the transitions which require large amount of energy but the intensity of the absorption is very high. However the conjugated double bonds require a small amount of energy for **$\pi \rightarrow \pi^*$ transitions.** As the no. of conjugated double increases, the absorption takes place at longer wave length.

In the study of organic compounds we are particularly interested in

i) $n \rightarrow \pi^*$ (π antibonding)

ii) $\pi \rightarrow \pi^*$ (π antibonding

Chromophor and auxochrome: chromophore is defined as a group which absorbs electromagnetic radiation in the UV or visible range.

Some important chromophores are >C = C<, λ_{max} = 175 nm $\pi \rightarrow \pi^*$ ε_{max} =15000

-C ≡ C- , λ_{max} = 175 nm $\pi \rightarrow \pi^*$ ε_{max} = 10000

>C = O λ_{max} = 180 nm $\pi \rightarrow \pi^*$ ε_{max} = 10000

λ_{max} = 160 nm $n \rightarrow \pi^*$ ε_{max} = 18000

λ_{max} = 285 nm $n \rightarrow \pi^*$ ε_{max} = 15

Auxochrome is a group which when attached to chromophore shifts the absorption maximum towards longer wavelengths along with an increase in the intensity of absorption . by itself it does not absorb any radiation. Some prominent auxochromic groups are –OH, $-NH_2$, -OR etc. When auxochrome $-NH_2$ is attached to benzene ring the absorption changes from 255 nm to 280 nm.

Bathochromic or red shift blue shift ; axochromes bring about an increase in the value of λ_{max} . the absorption is shifted towards higher wavelength or towards red portion of the spectrum .;The shift of absorption of light towards higher wavelength is known as **bathochromic or red shift**. This effect is also produced if two or more chromophores are present in conjugation. For eg. ethylene shows $\pi \rightarrow \pi^*$ transition at 170 nm whereas 1,3 butadiene shows absorption at 217 nm.

ii) Hypsochromic or blue shift: if we remove conjugation from a system, the absorption maximum is shifted towards lower wavelength or towards blue portion of the spectrum. This shifting of absorption maximum towards shorter wavelength by removing conjugation from a system is called hypsochromic shift or blue shift. For example. Protonation of aniline causes a

blue shift from 280 nm to 203 nm because the aniline ion has no electrons to participate in conjugation with benzene ring.

Hyperchromic shift: Introduction of an auxochrome into a system brings about an increase in the intensity of absorption. This effect is known as hyperchromic effect.

Conjugation effect:

Extended conjugation shifts the λ_{max} value towards longer wavelength (towards the red end of the spectrum. It also increases the intensity of the absorption.

For example: absorption of aromatic compounds move to longer wavelengths with size of conjugated system.

Among the benzene (λ_{max} 202 nm) , naphthalene (λ_{max} 275 nm), anthracene (λ_{max} 375 nm) and naphthacene (λ_{max} 473 nm), naphthacene has highest λ_{max} value.

λ_{max} values for a few functional groups

Compound and functional group	**λ_{max} value**
$CH_2 = CH_2$ -C=C-	171
HC≡ CH -C ≡ C-	187
$CH_2 = CH – CH = CH_2$, -C = C – C = C -	217
$(CH_3)_2$ C = O	166 & 189 & 285
Acetophenone	245.5
Benzene	203
Toluene	208
Aniline	230
CH_3 –CH = CH- CH =O	218

Applications of UV spectrum:

i) by the help of UV spectrum, it is possible to detect certain functional groups in the molecule.

ii) With the help of UV spectrum, we can establish the presence of conjugation in a molecule.

Problems:

1. which of the following compounds absorbs UV spectrum and which don't absorb.

Ethyl alcohol, ethane, propane, ethylene, methylamine, aniline, benzene, toluene, acetone, acetophenone,

The following compounds does not shows absorption
Ethyl alcohol, ethane, propane, methylamine

Rest of the following compounds shows absorption.

2. How will you differentiate between the following compounds using UVspectroscopy?

CH_3COCH_3 and $C_6H_5COCH_3$

$C_6H_5COCH_3$ shows absorption maximam at a much higher wave length due to the conjugation between the double bonds of the ring and C = O bond whereas in CH_3COCH_3 there is no conjugation. Therefore it absorbs at very low wave length that is at 186 nm.

3. The UV spectrum of acetone shows two important peaks at $\underline{\boldsymbol{\lambda_{max}}}$ 279 nm (ε_{max} 15) and $\underline{\boldsymbol{\lambda_{max}}}$ 198 nm (ε_{max} 900) . Identify the electronic transition for each.

$\underline{\boldsymbol{\lambda_{max}}}$ 279 nm (ε_{max} 15) corresponds to $n \rightarrow \pi^*$ transition.
$\underline{\lambda_{max}}$ 198 nm (ε_{max} 900) corresponds to $\pi \rightarrow \pi^*$ transitions

Q4. Arrange the following in increasing order of UV absorption spectrum.
i) anthracene ii) ethylene iii) naphthalene iv) butadiene

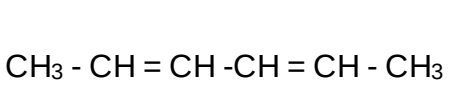
UV absorption maxima depends on the no. of conjugated double bonds in the compounds. Greater the no. of such double bonds greater will be the absorption maxima.

Above compounds are in the following increasing order of UV absorption.
Ethylene < butadiene < naphthalene < anthracene

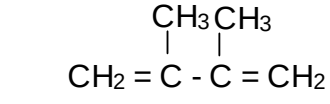
Woodward's rules for dienes :

Woodward, in 1942, reported a series of empirical rules for predicting the absorption maxima (λ_{max}) of dienes based on their substitution, conjugation etc.
These rules are
1. The basic λ_{max} of conjugated dienes is 217 nm
2. Add 5 nm for each alkyl substituent or a ring residue on the doubly bonded carbon.
3. Add 5 nm for each exocyclic double bond if any.

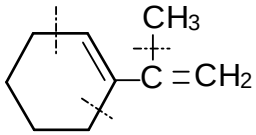
$CH_2=CH-CH=CH_2$	$CH_3 - CH=CH-CH=CH_2$
Calculated λ_{max} = 217 nm	217 + 5 = 222 nm
Observed λ_{max} = 217 nm	223 nm



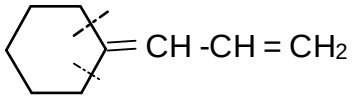
217 + 2 x 5 = 227 nm
(Obs.) 227 nm



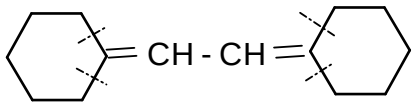
217 + 2 x 5 = 227 nm
Obs. 226 nm



217 + 3 x5 = 232 nm
Obs. 235 nm



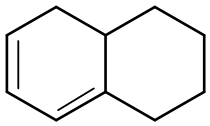
217 + 2 x5 + 1 x 5 = 232 nm
Obs 236.5 nm



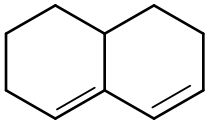
217 + 4 x 5 + 2 x 5 = 247 nm
Obs 248 nm

Woodward and Fieser rules for alicyclic dienes and Polyenes

The alicyclic diene base may be homoannular or heteroannular depending on whether the double bonds in the same ring or in different rings.



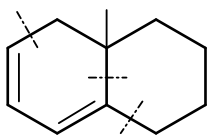
Homoannular diene



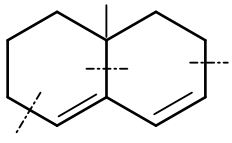
Heteroannular diene

The relevant rules are

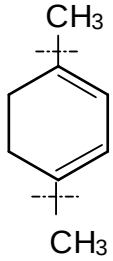
1. For heteroannular diene the basic λ_{max} is 214 nm but for homoannular diene the basic λ_{max} 253 nm . If both the homo annular and heteroannular dienes are present in a system, the basic λ_{max} of the homoannular diene is taken . i.e. 253 nm.
2. Add 5 nm for each alkyl substituent or ring residue
3. Add 30 nm for each additional double bond extending conjugation.
4 Add 5 nm for each exocyclic double bond.



253 + 3 x 5 + 5 = 275 nm
obs: 274 nm



214 + 3 x 5 + 5 = 234 nm
Obs. 234 nm



253 + 2 x 5 = 263 nm
265 nm

INFRARED SPECTROSCOPY

The atoms in a molecule are in a state of constant vibratory and rotatory motion. Thus, there are a number of vibrational and rotational changes taking place in the molecule.

On absorption of light, the molecules are excited from lower vibrational and rotational levels to higher ones. For affecting such transitions, frequencies in the infrared range are good enough. Molecules will absorb such frequencies as are needed to excite molecules from lower vibrational levels to permitted higher levels. Every bond and every functional group has a specific absorption frequency. A graph between the amount of absorption or emission of infrared light against the frequency or wavelength of the light is called infrared spectrum.

IR region has the range of 0.8 to 20 μm or 800 to 20000 nm. The region between 0.8 to 2.5 μm is called the near infrared region and the one between 15 to 20 μm is called far infrared region. The region between 2.5 to 15 μm is the proper infrared region. Generally, wave no. ($\bar{\nu}$) which is the no. of vibrations of the radiation per centimeter is used for describing IR spectra. The unit of wave no is cm^{-1}. The other unit which is used in connection with IR spectra is μm. the relation between μm and cm^{-1} can be calculated as given below.

1 μm. = 10^{-6} m = 10^{-4} cm

If λ is given in μm , it can be converted into cm by multiplying by 10^{-4}

Thus

$$\bar{\nu} \text{ (wave no).} = \frac{1}{\lambda \times 10^{-4}} = \frac{10^4}{\lambda} \text{ cm}^{-1}$$

Major bands in the IR spectra of different classes of compounds:

a) Alkanes: Mainly the C-H stretching bands are around 2900 cm^{-1} and C – C stretching band are around 1400 cm^{-1} are observed.

b) Alkenes: The C-H stretching bands in alkenes appear at slightly higher frequency (around 3000 cm^{-1} than in alkanes.

The C = C stretching band in alkenes is also at a higher frequency (1600- 1700 cm^{-1}) than in alkanes.

c) Alkynes:

C-H stretching band in alkynes is at still higher frequency (3310-3270 cm^{-1} than in alkanes.

The C ≡ C stretching band in alkynes appears at 2250 – 2150 cm^{-1} depending on the location of triple bond.

d) Alcohols: they show broad band at around 3650 – 3500 cm^{-1} corresponding to O – H stretching frequency.

They also show strong C – O stretching band at 1050 – 1200 cm^{-1}

e) Ketones: they show a strong sharp carbonyl stretching band around 1700 cm^{-1} . there are however, characteristic shifts in this band depending on the structural environment of the carbonyl group.

f) Aldehydes: they show the carbonyl band at about 10-15 cm^{-1} above the corresponding ketones. i.e. around 1720 cm^{-1}. In addition, aldehydes show characteristic C – H stretching bands at around 3000 cm^{-1}

g) Esters: The carbonyl band in esters appears at higher frequency than that observed in ketones. This is at around 1750 cm^{-1}

h) Amines : Primary and secondary amines but not the tertiary amines show the N – H stretching band at 3500 – 3300 cm^{-1} and N – H deformation band at 1650 – 1500 cm^{-1} .

i) Amides: amides show similar band as in amines. In addition, they show a carbonyl band at 1710 – 1610 cm^{-1}

IR absorption bands of some important bonds and functional groups:

Type of compound	Bond	Frequency (in cm^{-1})
Alkanes	C-H	2850-2960
Alkenes	=C-H C = C	3010-3100 1620-1680
Alkynes	≡ C-H C ≡ C	3300 2100-2600
Aromatic ring	C-H C-C	3000-3100 1510-1600
Aldehydes	C-H	2650-2880
Aldehydes, ketones and carboxylic acids, esters	C =O	1680-1760
Alcohols & phenols	O-H	3400-3650
Amines	N-H	3300-3500

Problems; l

Q 1. IR spectrum of acetone gives two maxima due to C-H vibrations at 1360 cm^{-1} and 3000 cm^{-1}

Identify the stretching and bending mode.

Bending vibrations require less energy than stretching vibrations. Hence absorption of IR in case of bending vibration will occur at longer wavelength or lower frequency than in case of stretching vibrations.

i) The absorption frequency of 1360 cm^{-1} is due to C – H bending vibrations.
ii) The absorption frequency of 3000 cm^{-1} is due to C-H stretching vibrations

Q2. What absorption of IR spectra could be used to distinguish between acetone and ethanol?

Acetone gives a band at 1700 cm^{-1} due to C = O stretching whereas ethanol gives a band at 3400 cm^{-1} due to O – H stretching (hydrogen bonded)

Q3. How will you distinguish acetic acid and acetone using IR spectroscopy?

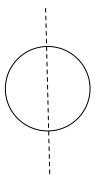
Both shows absorption bands at about 1700 cm^{-1} due C = O stretching. In addition to this spectrum of acetic acid shows absorption band between 2500 – 3000 cm^{-1} due to hydrogen bonded O –H group of carboxyl group which is not observed in the spectrum of acetone.

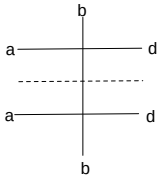
STEREO CHEMISTRY OF ORGANIC COMPOUNDS

Elements of symmetry & asymmetry:

i) Plane of symmetry (σ): A plane of symmetry is a plane which divides an object in such a way that the part of it on one side of the plane is the mirror image of that on the other side of the plane.

For example: following objects are having plane of symmetry.



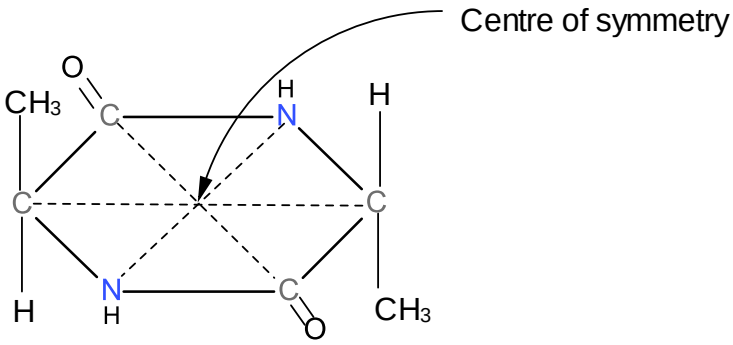


A molecule is symmetric if it has plane of symmetry and asymmetric if it has no plane of symmetry.

ii) A centre of symmetry or centre of inversion (i):

A centre of symmetry is a point from which line is drawn on one side and produced an equal distance on the other side will meet identical point in the molecule.

This can be applied particularly to those of ring systems.



iii) Axis of symmetry: An axis of symmetry is an axis about which the molecule can be rotated by $360^o/n$ and thereby produce a molecule indistinguishable from the original molecule (rotation is usually taken in clockwise)

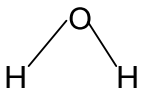
All linear molecules have a C_{oo} axis. An equivalent configuration is always obtained whatever is the angle of rotation.

Benzene possesses a C_6 axis perpendicular to the plane of the ring. It also has six C_2 axes in the plane of the ring.

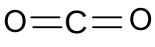
When the value of n is 1, then the axis is C_1 axis. The C_1 symmetry operation is carried out by rotating the molecule through 360°. The resultant molecule is a molecule identical with the original molecule. C_1 axis is said to be a trivial axis.

All molecules possess trivial axis.

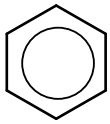
Ethylene has 3 C_2 axes.



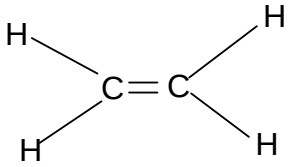
C_2 axis



C_{∞} axis



C_6 & $6C_2$ axis



3 C_2 axis

To find out whether the molecule is symmetric or not, it is required to determine whether the molecule possess i) a plane of symmetry ii) a centre of symmetry iii) an axis of symmetry

If the molecule contains at least one of these elements of symmetry, then the molecule is symmetric. If none of these elements of symmetry are present, then the molecule is asymmetric. Asymmetric molecule is not super imposable on its mirror image. It exhibits optical activity.

Isomerism: Two or more compounds have the same molecular formula but differ in the physical or chemical properties are called isomers & the phenomenon is called isomerism.

There are 2 main types of isomerism i) structural isomerism ii) stereo isomerism.

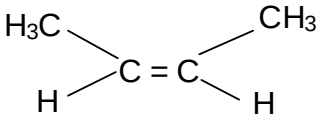
i) Structural isomerism: when the isomerism is due to difference in the arrangement of atoms within the molecule without any reference to space is called structural isomerism. In other words, structural isomers are compounds that have the same molecular formula but different structural formula.

ii) Stereo isomerism: when isomerism is caused by different arrangement of atoms or groups in space, the phenomenon is called stereo isomerism.

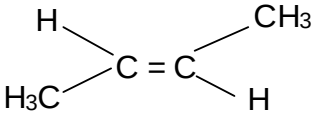
The stereo isomers have the same structural formula but differ in arrangement of atoms in space. In other words, stereo isomerism is exhibited by such compounds which have the same

structural formula but differ in configuration. [The term configuration refers to the three-dimensional arrangement of atoms that characterizes a particular compound.
They are of two types a) geometrical or cis-trans isomerism b) optical isomerism.
a) Geometrical isomerism or cis- trans isomerism:

Geometrical isomerism or cis-trans isomerism results from a restriction in rotation about double bonds or about single bonds in cyclic compounds.
The cis isomer is one in which 2 similar groups are on the same side of the double bond. The trans isomer is that in which 2 similar groups are on the opposite sides of the double bond. The restriction of rotation about the C-C double bond is responsible for geometrical isomerism in alkenes.
Examples: 1) 2-butene exhibit geometrical isomerism
The cis & trans isomers of 2-butene are shown below.

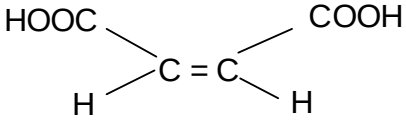


cis-2-butene (b.p.= 4°C)

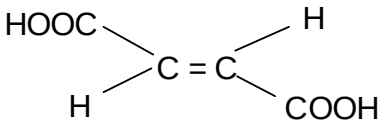


trans -2-butene (b.p.= 1°C)

2) Geometrical isomers of 2-butene dioic acid are as follows



Maleic acid (cis)
(m.p. = 288°C)

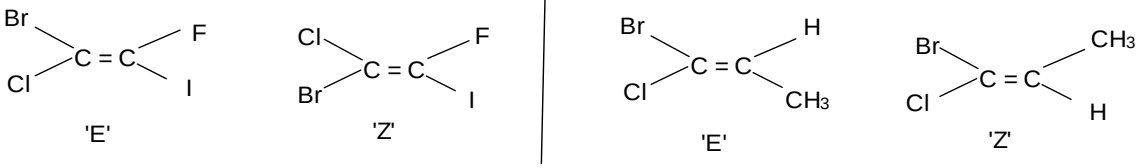


Fumaric acid (trans)
(m.p. = 130°C)

The trans isomers are more stable than the corresponding cis isomers. This is because in the cis isomer, the bulky groups are on the same side of the double bond. The stearic repulsion of the bulky groups makes the cis isomer less stable than the trans isomer in which the bulky groups are far apart [they are on the opposite sides of the double bond]

E,Z notation for geometrical isomerism:

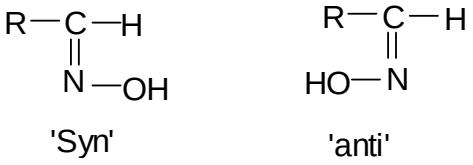
Cis isomer is one in which 2 identical groups are on the same side of the double bond and in trans isomer the two identical groups are on opposite side. When there are three or four different groups attached to double bond carbons, it is difficult to assign cis or trans designation to such isomers. Thus a new system of isomer assignment has been adopted in place of the older cis-trans system. This is called E and Z system. In this method, first determine the group of highest priority on each carbon (an atom with higher atomic number has higher priority). If the two highest priority groups are together on the same side, the configuration is Z (Cerman: Zusammen = together). If they are on opposite sides, the configuration is E (German, Entgegen = opposite)



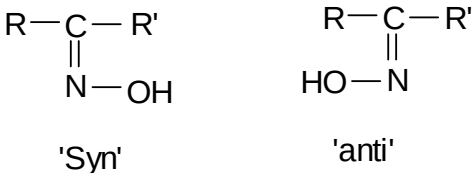
Geometrical isomerism of oximes:

Oximes are capable of exhibiting geometrical isomerism. Stereo isomers of oximes are designated as syn (equivalent to cis) and anti (equivalent to trans)

In the case of aldoxime, the syn isomer is one in which H and OH groups are on the same side of the double bond and in anti isomer they are on the opposite side.

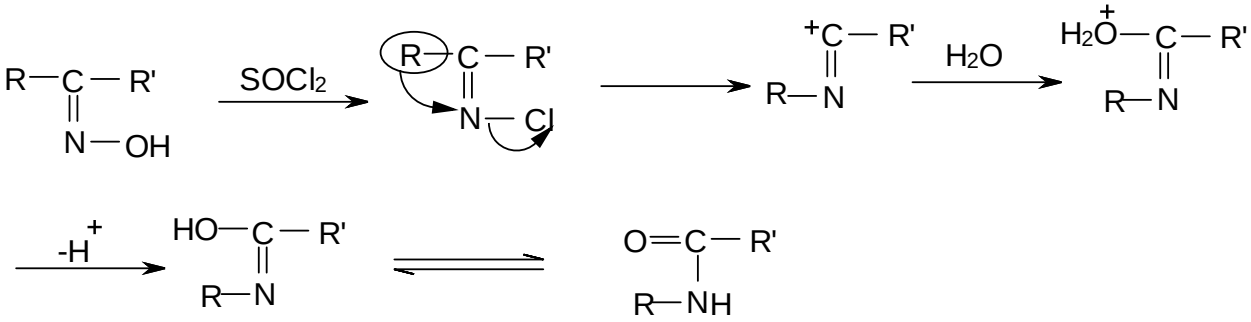


The ketoximes of mixed ketones, R-CO-R^1 also shows geometrical isomerism

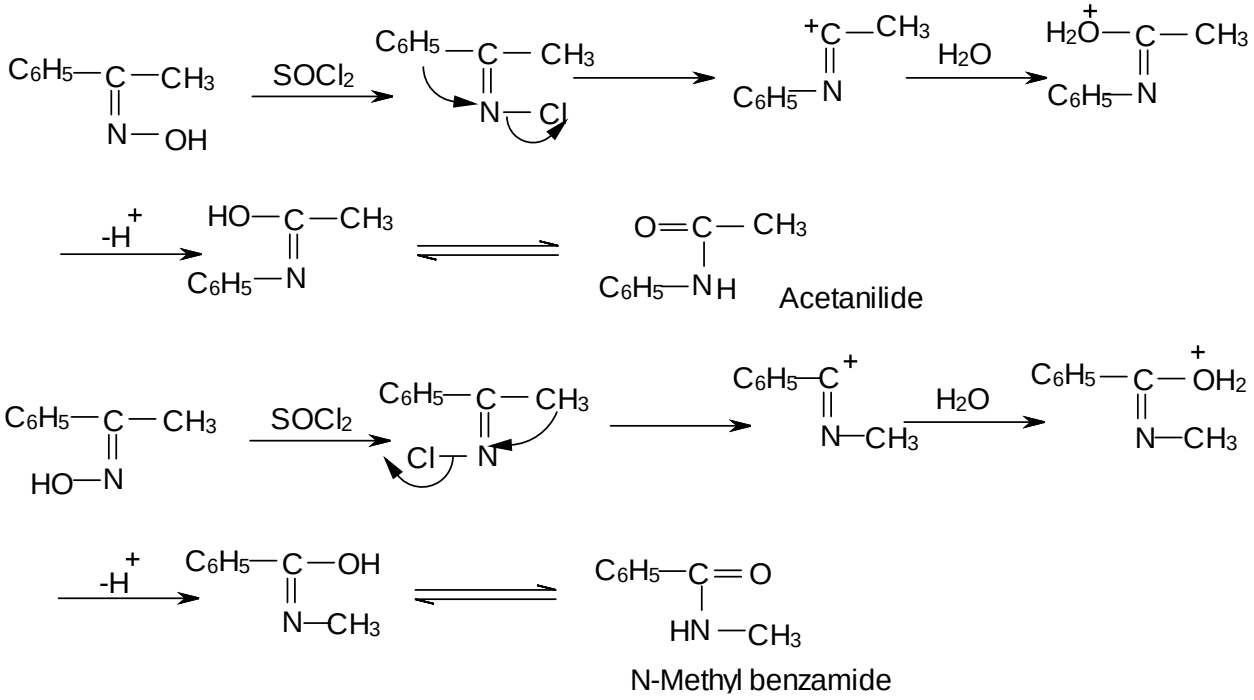


Geometrical isomerism is not possible in ketoximes derived from simple ketone in which R and R^1 are same.

Beckmann rearrangement: a ketoxime is treated with an acidic reagent such as $SOCl_2$ or phosphoric acid or sulphuric acid., when it undergoes molecular rearrangement to give the acid amide. During the rearrangement always trans hydrocarbon group R (with respect to hydroxyl group attached to N-atom) migrates to give the amide.

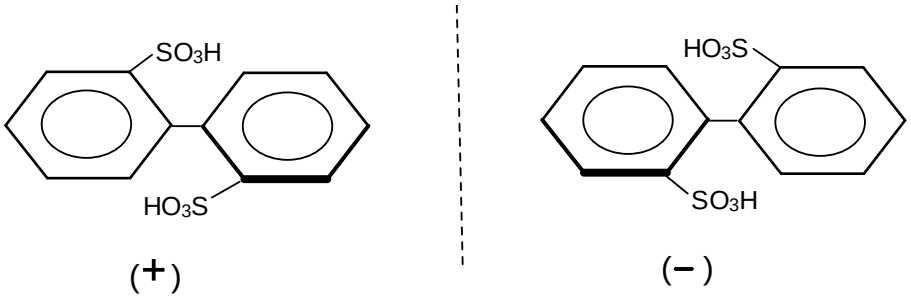


Determination of configuration of oxime : Beckmann rearrangement of syn-methyl-phenyl oxime gives acetanilide whereas anti-methyl-phenyl oxime gives N-methylbenzamide. By identifying the product, the configuration of ketoxime can be determined.



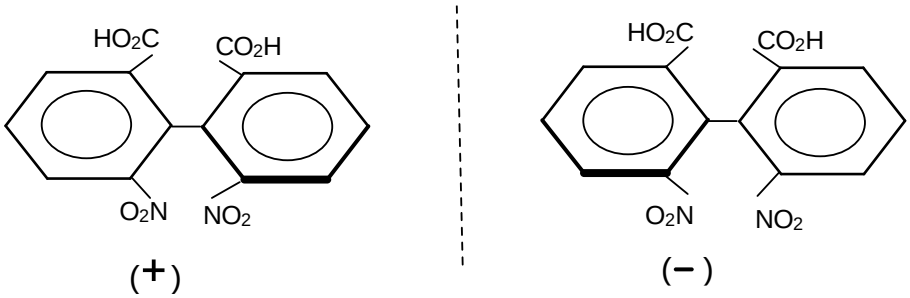
Optical activity of biphenyl derivatives

If substituents in the 2-positions are large enough to prevent rotation about the bond joining the two benzene rings then such biphenyls shows optical activity.
For example: Biphenyl-2,2[1]-disulphonic acid exists in 2 forms which are mirror images of each other.



The two forms are non super imposable mirror images.
Due to the presence of bulky o,o' substituents in biphenyl compounds, C-C bond between 2 phenyl groups is prevented from rotation. The two phenyl rings will remain co-axial but not co planar which gives rise to molecular dissymmetry. Therefore biphenyls with bulky o,o' – substituents exhibit optical activity.

e. g: 2 2,2'- dinitro diphenic acid is optically active and exist in two forms which are mirror images of each other.



Biphenyl without any substituents is optically inactive

OPTICAL ISOMERISM:

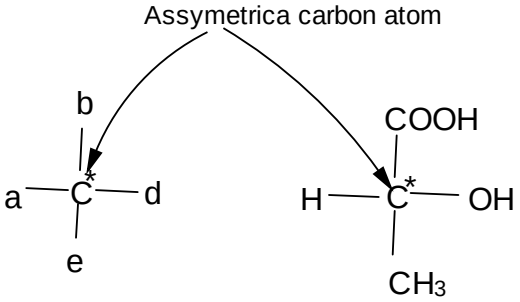
Optical isomerism is a type of sterio isomerism. An optically active compound can exist in 2 isomeric forms which rotate the plane polarized light in opposite directions. These are called optical isomers & the phenomenon is called optical isomerism. The isomer which rotates the plane of polarized light to the right (clockwise direction) is known as dextro rotatory isomer. It is indicated by sign(+). The isomer which rotates the plane of polarized light to the left (anticlockwise direction) is known as laevo rotatory isomer. It is indicated by the sign (-). The optical rotatory powers of two isomers are equal in magnitude but opposite in sign.

Optical isomers have the same physical properties (m.p., b.p., density etc.). They have the same specific rotation but with opposite sign.

Chirality:

The molecules, which are having no elements of symmetry (e.g. plane of symmetry or centre of symmetry) are called asymmetric molecules or chiral molecules. An asymmetric carbon is called chiral carbon. A symmetric object is to be referred as achiral.

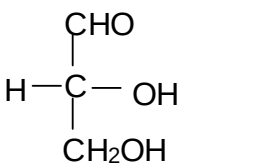
Asymmetric carbon atom: a carbon atom which is bonded to four different groups is called asymmetric carbon atom.



An asymmetric carbon in formula is usually indicated by an asterisk (*) written near it. The molecule having asymmetric carbon atom lacks plane of symmetry. Such a molecule is called asymmetric molecule. Dissymmetric or chiral term is also used for asymmetric molecule.

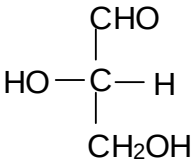
D & L configuration:

D – L system is used to specify the configuration at the asymmetric carbon atom. In this system, the configuration of an enantiomer is related to a standard, glyceraldehyde. The two forms of gylceraldehyde were arbitrarily assigned the absolute configurations as shown below.



(+) glyceraldehyde

D-configuration

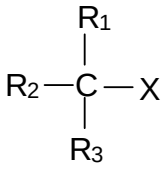


(-) glyceraldehyde

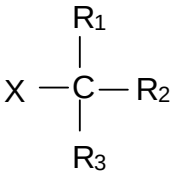
L configuration

If the configuration at the asymmetric carbon atom of a compound can be related to D (+) glyceraldehydes, it belongs to D- series. And it can be related to L (-) – glyceraldehydes, the compound belongs L – series.

In general, the absolute configuration of a substituent 'X' at the asymmetric centre is specified by writing the projection formula with the carbon chain vertical with the lowest numbered carbon at the top. The D configuration is one that has the substituent X on the bond extending to the right of the asymmetric carbon, whereas the L configuration has the substituent X on the left as shown below.



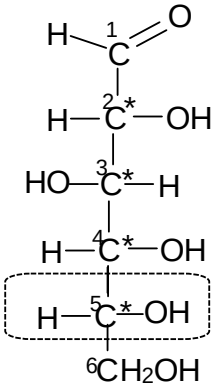
D-configuration



L configuration

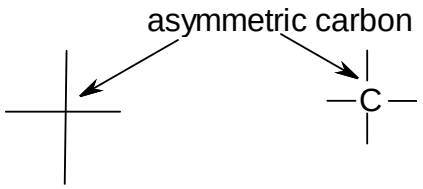
If there are several asymmetric carbon atoms in a molecule as in glucose, then the configuration of highest numbered asymmetric carbon is referred to glyceraldehyde to determine the overall configuration of the molecule.

For naturally occurring glucose, highest numbered carbon is C-5 on which -OH is to the right. Hence it belongs to D-series.

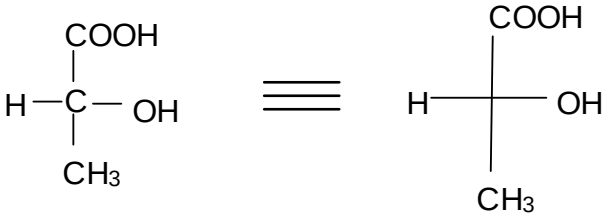


D- glucose (open chain)

Fischer projection formula.
Fischer projections provide an easy way to draw three – dimensional molecules on paper.
Fischer rules for showing the arrangement around an asymmetric carbon are as follows.
1. The carbon chain of the compound is arranged vertically with the most oxidized carbon or carbon no. 1 at the top.
2. Represent the asymmetric carbon (s) at the intersection of crossed lines.



The Fischer projection of one of the enantiomers of lactic acid is shown below.

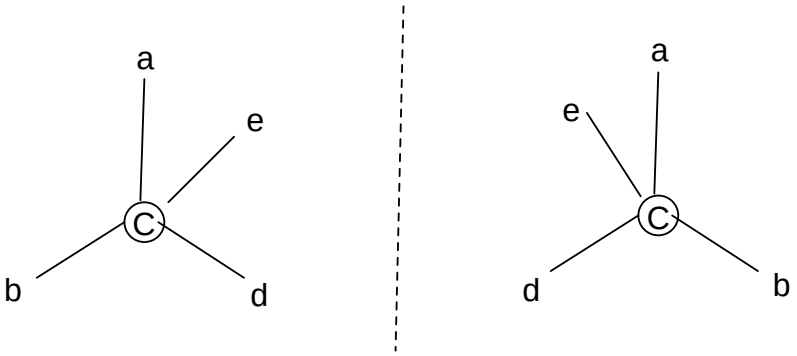


Fischer projection of (+) lactic acid

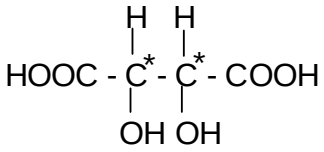
Racemic mixture: An equimolar mixture of two optical isomers (dextro & laevo) will not rotate the plane of polarized light at all & is said to be racemic mixture.

Chirality or molecular dissymmetry is the cause of optical isomerism.

The necessary condition for a molecule to exhibit optical isomerism is molecular dissymmetry or chirality. Thus all organic compounds which contain an asymmetric carbon atom are chiral or dissymmetric & exist in two tetrahedral forms. These two forms are related to each other as an object to its mirror image & are non super imposable. The two models of structures I & II stands for dextro & laevo. They are related to each other as object to mirror image. They are commonly called enantiomorphs or enantiomers.



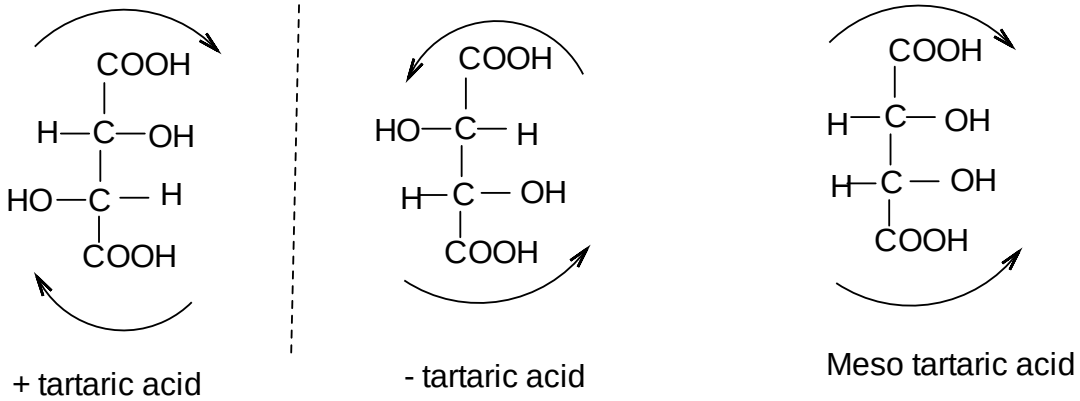
Optical isomerism in tartaric acid: Tartaric acid contains two asymmetric carbon atoms.



tartaric acid

Tartaric acid has four forms of isomers. Two of them are optically active & two are optically inactive.

The optically active forms are related to each other as an object to mirror image i. e. they are enantiomers.



+ tartaric acid

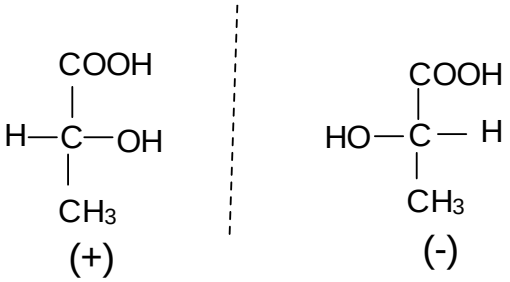
- tartaric acid

Meso tartaric acid

1) (+) tartaric acid: It rotates the plane of polarized light to the right & is called dextro rotatory.
2) (-) tartaric acid: it rotates the plane of polarized light to left (anticlockwise) direction. & is called laevo rotatory. (+) tartaric acid is
the mirror image of the (-) tartaric acid & vice versa.
3) mesotartaric acid: it possess a plane of symmetry & is consequently optically inactive. Optical activity of one half of the molecule is internally compensated by the other half (i.e. optical rotation due to one asymmetric carbon atom is cancelled by that of the other)
4) Tartaric acid or racemic tartaric acid: it is the equimolar mixture of (+) & (-) forms. It does not rotate the plane of polarized light i.e. it is optically inactive. The optically rotation of one half of the mixture is externally compensated by other half .

Enantiomers: optical isomers that are mirror images of each other are called enantiomers. These always exist as discrete pairs.

For example: there are 2 optical isomers of lactic acid. One is the mirror image of the other. i.e. A is the mirror image of B . Enantiomers have identical properties in all respects except in their interaction with plane polarized light.



They have same melting point, density, solubility, colour & reactivity towards acids & bases etc. However they differ in the direction in which they rotate the plane of polarized light. Both rotate the plane of polarized light exactly to the same extent [same angle] but one rotates the plane to the right (clockwise) and is called dextro rotatory & the other rotates the plane to the left (anticlockwise) called laevo rotatory.

A mixture of equal amounts of two enantiomers is called a racemic mixuture. Such a mixture is optically inactive i.e. it does not rotate the plane of plarised light. (In fact the isomers rotate the plane of polarized light equally in opposite directions & hence cancel each other)

Diastereomers: Each asymmetric carbon atom in a molecule doubles the number of theoretically possible isomers. Hence molecules with 'n' asymmetric carbon atoms should have 2^n stereo isomers.

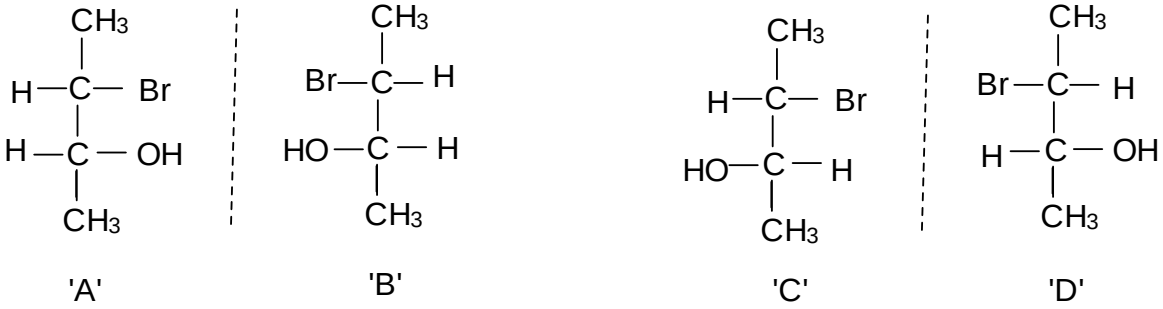


Figure shows the four isomers of 3-bromo-2-butanol which have 2 asymmetric carbon atoms. In the above figure (A) is the mirror image of (B) & (C) is the mirror image of (D). Thus A & B and C & D are two pairs of enantiomers.

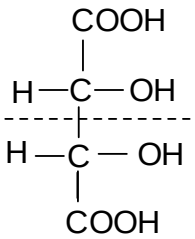
Now compare A with C. They are neither super imposable nor mirror images. They are called diasteriomers. Therefore stereomers that are not mirror images of each other are called diasteriomers. Thus in the above example A & C, B & C, B & D, A & D are the four pairs of diasteriomrers.

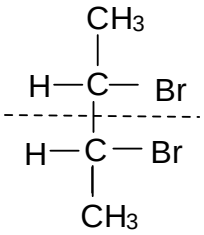
Diasteriomers have different properties. They have different m.p.,b.p. & solubility etc. They differ in chemical reactivity towards most reagents.

Meso compounds:

A compound with two or more asymmetric carbon atoms but also having a plane of symmetry is called the meso-compound. Fig. shows 2 meso compounds. Meso compounds have plane of symmetry dividing them midway between the two asymmetric carbons, so that one half

of the molecule is the mirror image of the other. Meso compounds are optically inactive. Optical activity of meso compound is internally compensated.





Racemisation:

Conversion of an optically active compound into a recemic mixture is called recemisation. (Racemic mixture is a mixture containing equal amounts of + and – isomers.) Racemisation of optically active compound can be accomplished by means of heat or light or chemical reagents. The process involves the change of one half of the optically active compound to the isomer of the opposite rotation thereby resulting in a racemic mixture.

2 (+) A ⟶ (+)A + (-) A

2 (-) A ⟶ (+)A + (-) A

Resolution of racemic mixture:

The separation of a racemic mixture into its two optically active components [+ & - isomers] is known as resolution

1. Chemical resolution: This is the most general method of resolution. In this method the racemic mixture is made to combine with another suitable optically active compound. The products thus obtained are no longer mirror image isomers (i.e. they are diastereomers) & may be separated by crystallization.

For example: a solution of racemic lactic acid is treated with an optically active base such as (-) brucine. The resulting products will consist of two salts namely

(-) brucine (+) lactate and (-)brucine (-) lactate.

i.e. salt of (+) acid & (-) base. ii) salt of (-) acid & (-) base.

These salts are not enantiomers therefore they have different solubility in water and can be separated by fractional crystallization. The isolated salts are then treated with dil H_2SO_4 when the optically active acids are regenerated. In a similar way, a recemic mixture of optically active forms of a base can be separated by the use of an optically active acid.

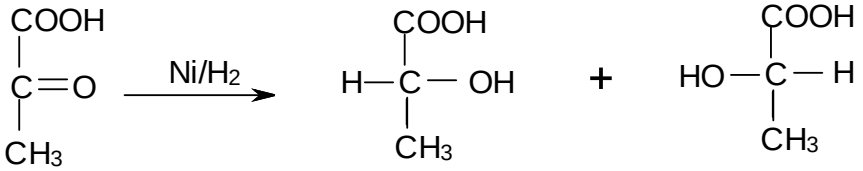
2) Biochemical method:

In this method the resolution is affected by destroying one form (+) or (-) by a biochemical process. When certain bacteria or moulds are added to a solution of racemic mixture, they bring about the decomposition of one of the optically active forms more rapidly than the other. If ordinary mould, penicillium glaucum is added to a solution of $(\pm)$ tartaric acid the solution becomes laevo rotatory due to the destruction of dextro form. This is because Penicillium glaucum decomposes (+) tartaric acid more readily than (-) isomer. The (-) form can be isolated from the solution by crystallization.

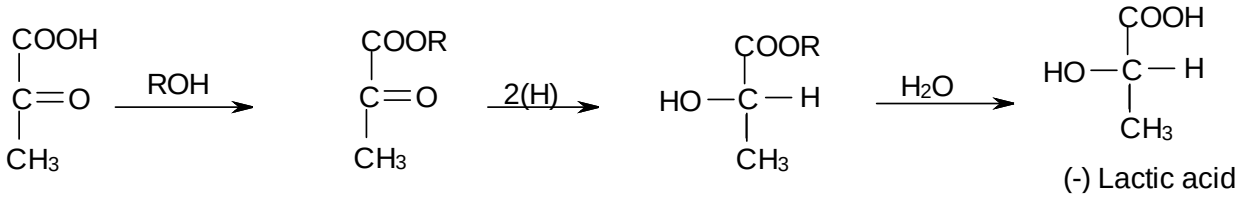
Asymmetric synthesis:

If a compound containing an asymmetric carbon atom is synthesized by ordinary laboratory method from a symmetric compound, the product will be racemic mixture. If however such a synthesis is carried out under the presence of a suitable optically active reagent, only one of the optically active isomers (+) or (-) is formed. This process in which an optically active compound is prepared from symmetric compound by the intermediate use of another optically active compound to yield (+) or (-) isomer directly, is termed as asymmetric synthesis or partial asymmetric synthesis.

e.g. when pyruvic acid is reduced in the laboratory leads to (±) lactic acid. However when pyruvic acid is first combined with an optically active alcohol such as (-) menthol, to form an ester which is then reduced and the product obtained upon hydrolysis yields (-) lactic acid in excess.



(+) Lactic acid (-) Lactic acid

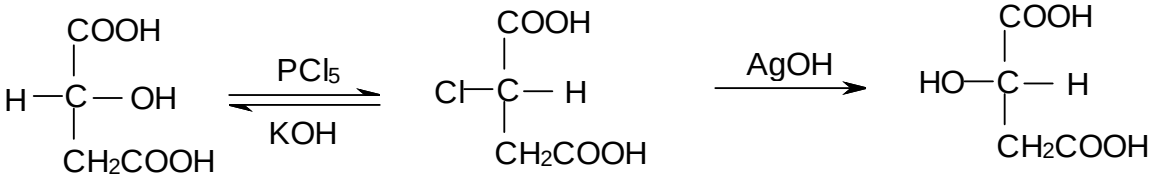


ROH = (-) menthol

(-) Lactic acid
(major product)

Absolute asymmetric synthesis: is a method of preparing optically active compound from racemic mixtures by using **circularly polarized light**. Here one of the enantiomer is decomposed or destroyed preferentially.

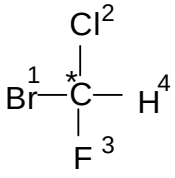
Walden inversion: When a group attached to an asymmetric carbon atom is replaced, the configuration of the new compound may be opposite to that of original. It is defined as the conversion of the (+) form into (-) form or vice versa without recourse to resolution.



(+) Malic acid (-) Chloro succinic acid (-) Malic acid

Specifying absolute configuration: R & S system

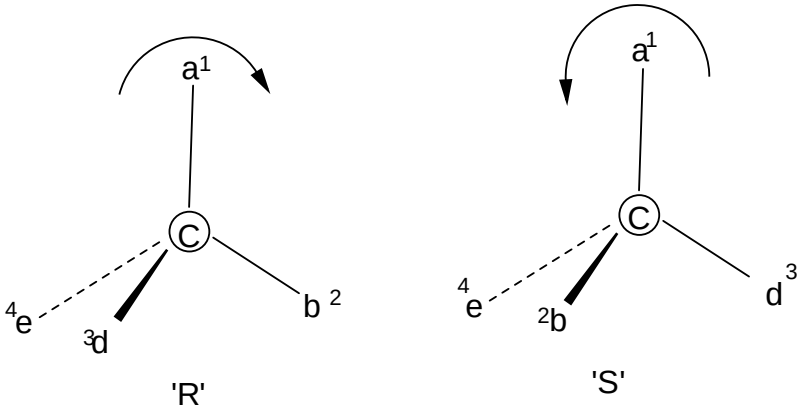
The actual three dimensional arrangement of groups in an asymmetric molecule is called its absolute configuration. We can specify the absolute configuration using R & S system. The R & S system is also called Cahn-Ingold-Prelog system named after its inventors.
In the R, S system the four groups attached to the asymmetric carbon atom are arranged in the decreasing order of priority.(1,2,3,4) by applying the following sequence rules.
Rule 1: of the atoms attached directly to the asymmetric carbon atom, the one with the highest



atomic number has the highest priority. For example:
I >Br > Cl > F > O > N > C > H

Rule 2: if two or more groups have identical first atoms attached to the asymmetric carbon atom, then determine priority order by considering atomic numbers of second atoms, and if the second atoms also identical the third atoms along the chain are examined.
For example: in methyl and ethyl the first atom is carbon and therefore atomic numbers of the second atoms H in methyl (at. No.1) and C in ethyl (at. No. 6) decide the priority order.
Therefore ethyl > methyl . while considering ethyl and n-propyl the second atom is also identical and hence the atoms (hydrogen & carbon) give the priority order.
i.e. n-propyl > ethyl
Rule 3: If the first atoms of the two groups have same substituents of higher atomic number the one with more substituents takes priority.
Thus $CHCl_2$ has a higher priority than CH_2Cl
Rule 4: a double bond is treated as though each atom of the double bond were bonded to two atoms.
Hence between groups –CHO & (O, O, H) and CH_2OH (O, H, H) the former will have higher priority.

The groups are arranged in the decreasing order of priority by applying above rules.
For example: in the tetrahedral model of the molecule below, if a has the highest priority (1) , b has the next higher (2), d has the next (3) and e has the lowest priority (4) then the molecule is viewed from the direction opposite to the lowest priority group.

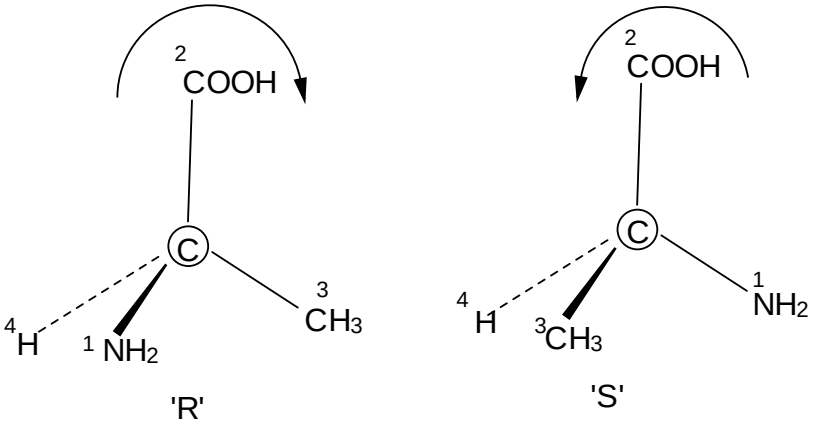


The conversion rule says that
i) if the eye while moving from a ⟶ b ⟶ d travels in a clockwise or right hand direction the configuration is R (Latin: Rectus = right)

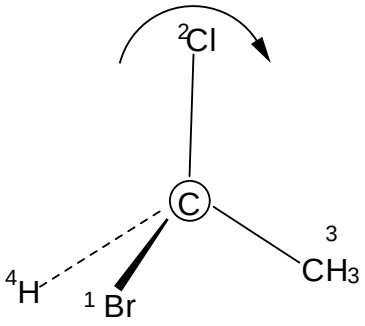
i) if the eye while moving from a ⟶ b ⟶ d travels in a counter clockwise or left hand direction the configuration is designated as S (Latin: Sinister = left

For example: the two enantiomers of 1,1- chlorobromoethane are written as follows. The four atoms which are directly bonded to asymmetric carbon in 1,1- chlorobromoethane are Br (at. No. 35), Cl (at. No. 17), C (at. No. 6) and H (at. No. 1)
Hence the priority order is Br > Cl > CH_3 > H
Now viewing the groups Br, Cl, CH_3 in tetrahedral structure, the R and S are assigned as follows.

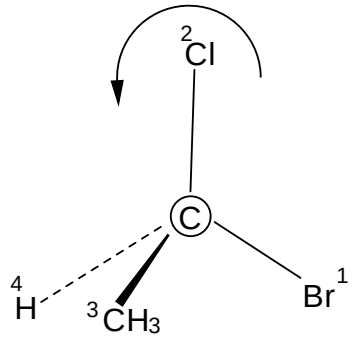
Example 2: Alanine



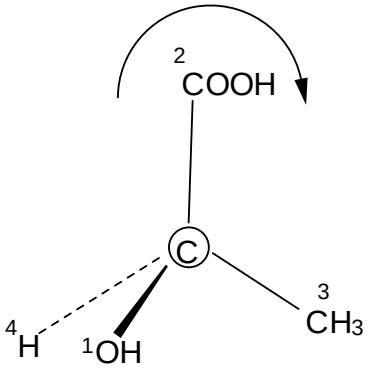
Example :1 Chlorobromoethane example 3. Lactic acid



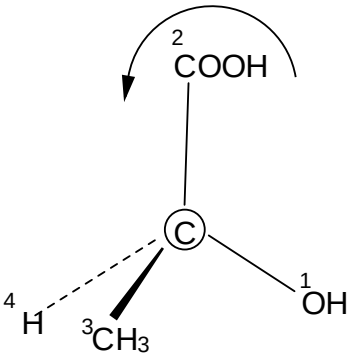
'R'



'S'

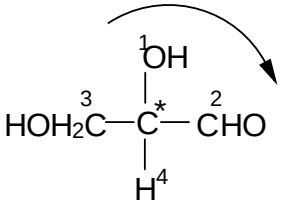


'R'



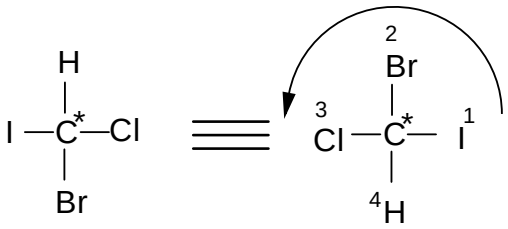
'S'

Configuration on the basis of Fischer Projection formula:
When a compound is represented by the Fischer projection formula, the configuration can be easily determined without constructing the model. To determine whether the eye travels clockwise or anticlockwise , we have to place the group or atom of the lowest priority at the bottom of the Fischer projection formula. The following four situations arise.
i) the atom/ group of lowest priority is at the bottom. In such case, simply rotate the eye in the order of decreasing priority.
For example: glyceraldehydes:



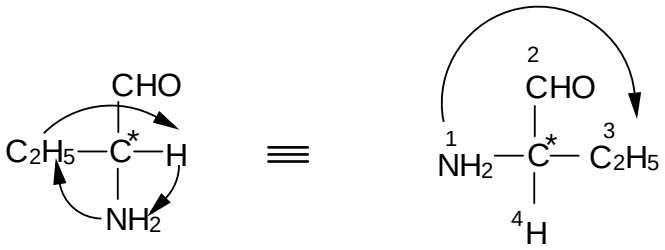
R-glyceraldehyde

ii) the atom/group of lowest priority is at the top. In such case rotate the molecule by 180° so as to bring the atom/group of lowest priority at the bottom. Then find the direction in the order of decreasing priority.
For example



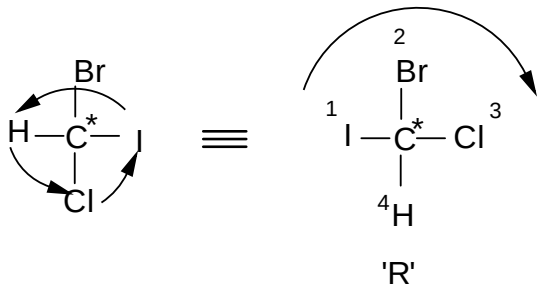
S-Bromochloroiodomethane

iii) the atom/group of lowest priority is at the right hand side of the horizontal line :
In such case without changing the position of atom/group at the top of the vertical end, change the position of other atoms/groups in the clockwise direction so that atom/group of lowest priority comes at the bottom. Then find the direction in the order of decreasing priority.



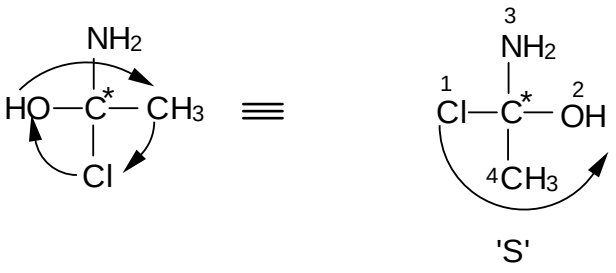
'R'

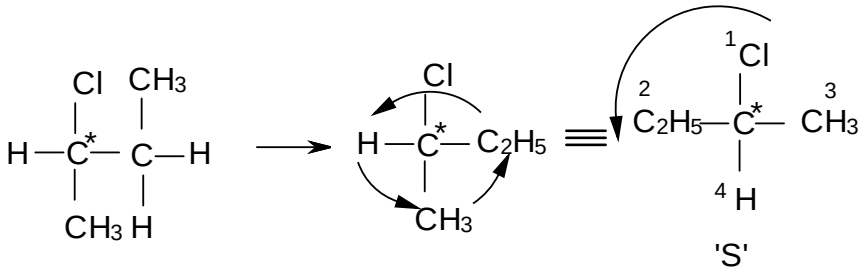
iv) the atom/group of lowest priority is at the left hand side of the horizontal line :
In such case without changing the position of atom/group at the top of the vertical end, change the position of other atoms/groups in the anti- clockwise direction so that atom/group of lowest priority comes at the bottom. Then find the direction in the order of decreasing priority.



'R'

Examples: Assign R& S configuration to the following





Conformation of alkanes:

We know from variety of evidence that groups bonded by a carbon- carbon single bonds are capable of free rotation.

The different arrangement of atoms in space that can be obtained by rotation of groups about C-C bond axis are called conformations or conformational isomers.

Alkanes can have infinite number of conformations by rotation around carbon-carbon single bonds.

For example, in ethane two tetrahedral methyl groups can rotate about the carbon carbon bond axis yielding several arrangements. Out of infinite number of possible conformations of ethane, two extreme arrangements are **Staggered and Eclipsed conformations.**

In the staggered conformation hydrogen atoms are as far apart as possible, whereas in the eclipsed conformation hydrogens are crowded together and almost eclipse each other.

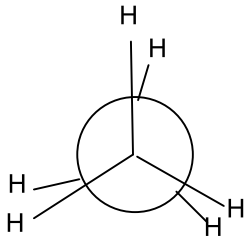
The infinite number of possible intermediate conformations are referred to as **Skew conformations.**

In Newman projection , one is viewing the C-C bond along the axis of connection. The front carbon atom is represented by the intersection of bonds from it, while the rear carbon appears as a circle.

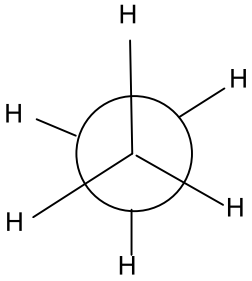


Front carbon Rear carbon

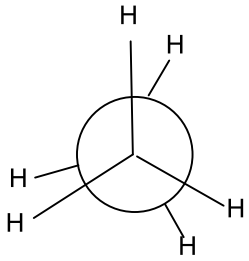
The Newman projection formulas for the conformations of ethane could be depicted as in figure.



Eclipsed



Staggered



Skew

(bonds of the rear carbon in the eclipsed form are shown slightly displaced simply to make them visible.)

The different conformations of ethane are not of equal stability. The staggered form in which the hydrogen atoms are perfectly staggered is the most stable conformation. Ethane has six carbon-hydrogen bonds attached to the central carbon-carbon single bond. In the staggered form, the electron clouds of these carbon –hydrogen bonds are as far apart as possible. Therefore staggered conformation of ethane is most stable and has least potential energy. Therefore staggered conformation is the dominant conformation of ethane molecule.

In the eclipsed conformation of ethane, the hydrogen atoms attached to each carbon are in direct opposition to each other. This permits minimum separation of the electrons of the six carbon-hydrogen bonds. The eclipsed conformation is, therefore, of highest energy and has the lowest stability.

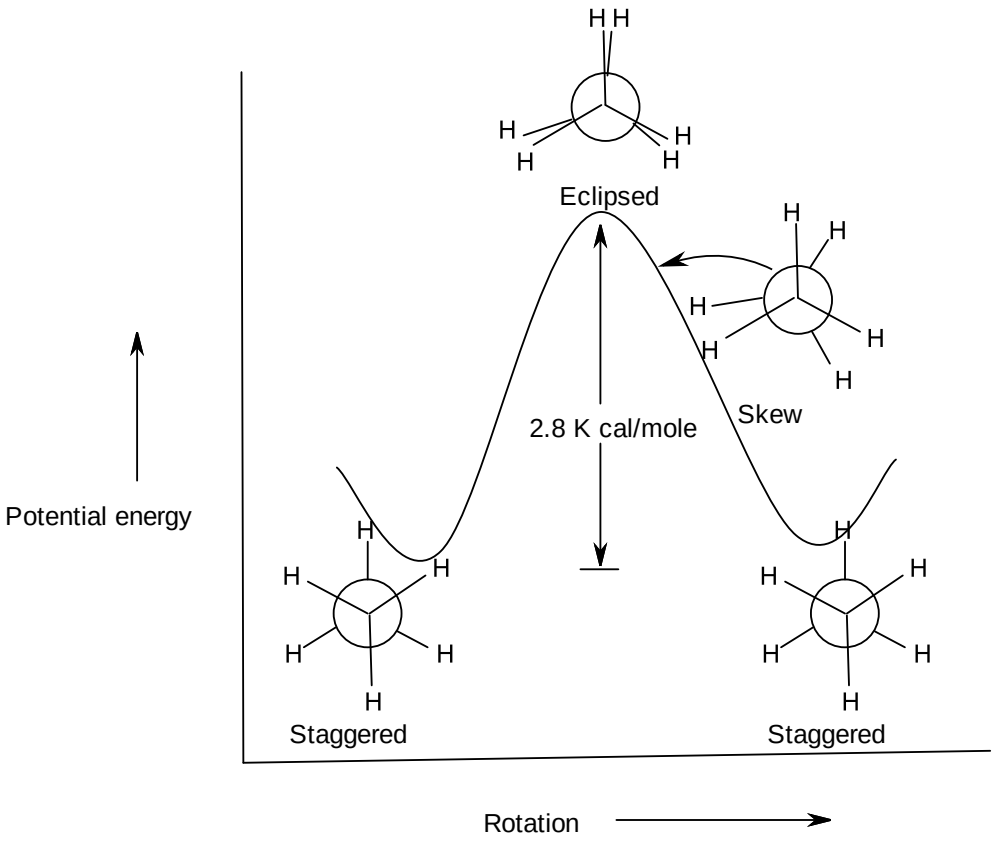
All the skew conformations of ethane will have stabilities in between that of staggered and eclipsed conformations. Hence the relative stabilities of the various conformations of ethane are

Staggered > skew > eclipsed

The staggered and eclipsed conformation of ethane can be regarded as rotational isomers, because they can be converted into one another by rotation about the C-C bond. A rotation of 60° converts a staggered conformation into an eclipsed conformation and vice versa.

Potential energy of ethane as function of degree of rotation about the C-C bond

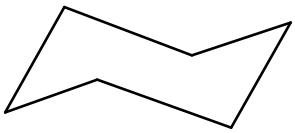
A plot of potential energy of an ethane molecule as a fuction of rotation about carbon – carbon bond is shown in fig. The staggered conformation has the minimum and eclipsed conformation has the maximum potential energy. The energy difference between the two extreme conformations is2.8 kcals/mole and it constitutes the energy barrier to rotation about the carbon-carbon bond. At room temperature, ethane molecules have more than enough energy to overcome this energy barrier of just 2.8 k cals/mole. In otherwords, rotation about the carbon – carbon single bond in ethane is almost free for all practical purposes.



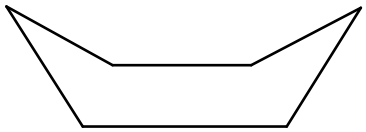
Potential energy of ethane as function of degree of rotation about the C-C bond

Conformations of cyclohexane:

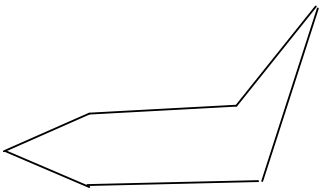
The cyclohexane ring can assume many shapes. A single cyclohexane molecule is in a continuous state of flexing or flipping into different shapes or conformations. Some of these shapes are shown below**.**



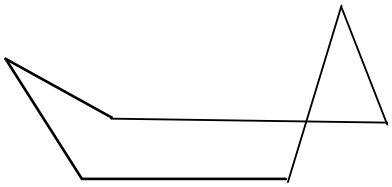
Chair form
(most stable)



boat form



Half chair

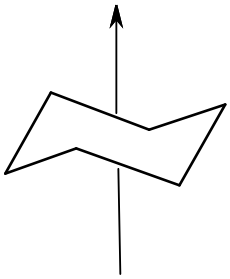


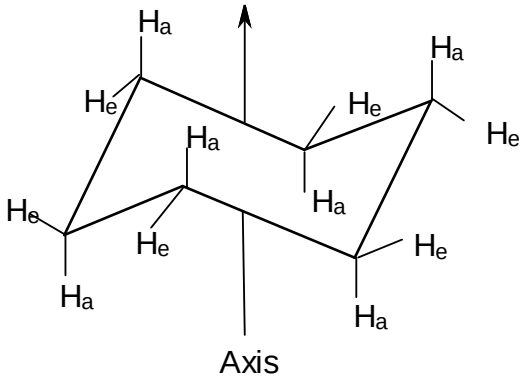
Twist boat

The conformations arise due to rotation around carbon carbon bonds. The chair form and the boat form are two extreme cases. Among the above conformations the chair form has the lowest energy, and half chair form has the highest energy. At any given time, most of the cyclohexane molecules are in the chair form.

Equatorial and axial hydrogens:

The carbon atom of the chair form of cyclohexane roughly lie on one plane and an axis can be drawn perpendicular to this plane.



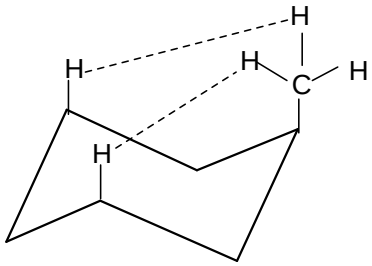


Each carbon atom of cyclohexane is bonded to two hydrogen atoms. The bond to one of these hydrogens lies in the rough plane of the ring & this hydrogen is called equatorial hydrogen. The bond to the other hydrogen atom is parallel to the axis. This hydrogen is called axial hydrogen.

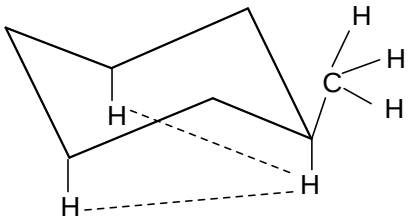
Conformations of mono substituted cyclohexanes:

When the methyl group in methyl cyclohexane is in the axial position, the methyl group and axial hydrogens present on the same side of the ring repel each other. This interaction between axial groups is called axial – axial interaction.

When the methyl group is in the equatorial position the repulsion is minimum. Therefore the energy of the conformation with an equatorial methyl group is lower and more stable.
At room temperature, about 95% of methyl cycohexane molecules are in the conformation in which the methyl group is equatorial.



Methyl cyclohexane molecule with methyl group at axial position (more repulsion) (less stable)



Methyl cyclohexane molecule with methyl group at equatorial position (Less repulsion) (more stable)